# EXPLICIT HOCHSCHILD ORBIFOLD DECOMPOSITION FOR SYMMETRIC POWERS OF A SMALL DG CATEGORY

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*Allowed standard prerequisites (editorial).*

- Automorphic tensor products, induction/base change and analytic Rankin-Selberg theory. Original complete bibliography and source-local auxiliary L-function arguments retained.

## 1. Introduction

This paper is a step towards developing "orbifold cohomology theory" for DG categories. It derives its inspiration from the computation of Hochschild and cyclic homology of twisted group rings (crossed product rings).

Our starting point is the following well-known result. Let $X = \operatorname{Spec} A$ be a smooth affine variety over an algebraically closed field $\Bbbk$ of characteristic 0. Let $G$ be a finite group acting on $X$ via a group homomorphism $\alpha : G \to Aut(X)$. Let $A_\alpha[G]$ be the corresponding twisted group ring. Its homology groups admit the following decomposition:

$$\mathrm{HC}_\bullet(A_\alpha[G], W) \simeq \Big(\bigoplus_{g\in G} \mathrm{HC}_\bullet(A(X^g), W)\Big)_G \tag{1.1}$$

Here $\mathrm{HC}_\bullet(\cdot, W)$ is the cyclic homology with coefficients in a module $W$: for different $W$ this gives Hochschild, cyclic and periodic cyclic homology [GJ93, §4]. Furthermore, $A(X^g)$ is the algebra of regular functions on the fixed point set $X^g$ and $\big(\dots\big)_G$ denotes taking the coinvariants of the $G$ action $g\colon X^h \to X^{ghg^{-1}}$.

The decomposition (1.1) has two steps. The first step is based on spectral sequence constructed by Feigin and Tsygan [FT87] for any $G$ acting on a unital algebra $A$ over a commutative ring $R$. For finite $G$ whose order is invertible in $R$, the spectral sequence reduces to a decomposition into a direct sum of the twists ${}_gA$ of the trivial $A$-$A$-bimodule by action of $g \in G$ [GJ93, §4]. For a smooth commutative $A$, the second step is the interpretation of these direct summands by Brylinski in terms of the fixed point sets of the $G$-action on $X = \operatorname{Spec} A$ [Bry87].

There are many geometrical generalisations of the decomposition (1.1): for smooth quasi-projective varieties over $\Bbbk$ [Bar03], as an orbifold version of the HKR isomorphism for a global quotient of a smooth variety [ACH19] and a recent version for derived Deligne-Mumford stacks [LS25].

Our aim is a noncommutative generalisation of (1.1) replacing $A$ by a smooth and proper DG category $\mathcal{A}$. We are guided by a noncommutative geometry perspective which views small DG categories considered up to Morita equivalence as noncommutative schemes [KS09, KKP07, Kal07, Orl16, Efi20]. The first step of the noncommutative decomposition (1.1) was established by Nordström for any small DG category $\mathcal{A}$ over a field $\Bbbk$ with $\mathrm{char}(\Bbbk) = 0$ [Nor25b]. The more delicate

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second step with its interpretation in terms of the fixed point loci, where we expect the smoothness of $\mathcal{A}$ to become necessary, remains open.

In this paper, we do it for the symmetric group $S_n$ acting on $\mathcal{A}^{\otimes n}$, the $n$-th tensor power of a DG category $\mathcal{A}$. The geometric fixed point loci are the diagonals $X^m \subset X^n$ which are easily interpreted as tensor powers $\mathcal{A}^{\otimes m}$. The conjugacy classes of $S_n$ are given by unordered partitions $\underline{\lambda} \vdash n$. Write $r_i(\underline{\lambda})$ for the number of parts of size $i$ in $\underline{\lambda}$ and $r(\underline{\lambda})$ for the total number of parts. The fixed point locus of $\underline{\lambda}$ is $\mathcal{A}^{\otimes r(\underline{\lambda})}$ and the centraliser acts on it as $S_{\underline{r}(\underline{\lambda})} := S_{r_1(\underline{\lambda})} \times \cdots \times S_{r_n(\underline{\lambda})}$. By the Künneth formula $\mathrm{HH}_\bullet(\mathcal{A}^{r(\underline{\lambda})})_{S_{\underline{r}(\underline{\lambda})}} \simeq \mathrm{Sym}^{\underline{r}(\underline{\lambda})} \mathrm{HH}_\bullet(\mathcal{A})$. Our first main result is:

**Theorem 1.1** (see Theorem 4.1)**.** *Let $\mathcal{A}$ be a small DG category and $n \geq 0$. The following compositions are mutually inverse isomorphisms:*

$$\mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A}) \underset{\xi}{\overset{\nu}{\rightleftarrows}} \bigoplus_{\underline{\lambda}\vdash n} \mathrm{HH}_\bullet(\mathcal{A}^{\otimes n};\sigma_{\underline{\lambda}})_{C(\sigma_{\underline{\lambda}})} \underset{\sum g_{\underline{\lambda}}}{\overset{\sum f_{\underline{\lambda}}}{\rightleftarrows}} \bigoplus_{\underline{\lambda}\vdash n} \mathrm{Sym}^{\underline{r}(\underline{\lambda})} \mathrm{HH}_\bullet(\mathcal{A}), \tag{1.2}$$

*where $\nu$ and $\xi$ are the maps explicitly constructed in §4.1 and $f_{\underline{\lambda}}$ and $g_{\underline{\lambda}}$ – in §4.2.*

The action of $S_n$ on $\mathcal{A}^{\otimes n}$ is strong, i.e. it acts by automorphisms, rather than autoequivalences. We can thus use the formalism of equivariant DG categories introduced in [GKL21, §4.8]. For any DG category $\mathcal{A}$ with a strong action of a finite group $G$, the equivariant category $\mathcal{A} \rtimes G$ is an analogue of the twisted group ring, see Defn. 2.1. There is an isomorphism $\mathbf{Mod}^G\text{-}\mathcal{A} \simeq \mathbf{Mod}\text{-}(\mathcal{A} \rtimes G)$, i.e. the $G$-equivariant $\mathcal{A}$-modules are the same as ordinary $(\mathcal{A} \rtimes G)$-modules. We can thus use $\mathcal{A} \rtimes S_n$ for $\mathrm{Sym}^n \mathcal{A}$, as oppposed to $h\mathcal{P}erf^{S_n}\text{-}\mathcal{A}$ in an earlier approach of Kapranov and Ganter [GK14].

In the Morita DG enhancement framework, both $\mathcal{A} \rtimes S_n$ and $h\mathcal{P}erf^{S_n}\text{-}\mathcal{A}$ correspond to the same enhanced triangulated category $\mathrm{Sym}^n \mathcal{A}$, the noncommutative analogue of the symmetric quotient stack $[X^n/S_n]$. The advantage is that $\mathcal{A} \rtimes G$ is a much smaller category, and many computations become simpler. In particular, the explicit isomorphisms $\nu$ and $\xi$ in (1.2) give in this simpler language the decomposition obtained by Nordström in [Nor25b] using the formalism of [GK14].

The decomposition of Theorem 1.2 implies that the total Hochschild homology $\bigoplus_{n\geq 0} \mathrm{HH}_\bullet(\mathrm{Sym}^n \mathcal{A})$ can be given the structure of a certain symmetric algebra, the result conjectured in [BFK23, Conj. 3.24] and [GKL21, Cor. 8.6] :

**Theorem 1.2** (see Theorem 4.2)**.** *Let $\mathcal{A}$ be a small DG category. We have explicit mutually inverse isomorphisms*

$$\bigoplus_{n\geq 0} \mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A}) \underset{\eta}{\overset{\zeta}{\underset{\simeq}{\rightleftarrows}}} S^*(\mathrm{HH}_\bullet(\mathcal{A}) \otimes t\Bbbk[t]), \tag{1.3}$$

*where the RHS is the graded symmetric algebra of $\mathrm{HH}_\bullet(\mathcal{A}) \otimes t\Bbbk[t]$ with $t\Bbbk[t] = \Bbbk t \oplus \Bbbk t^2 \oplus \ldots$ being of $\deg = 0$. Isomorphisms $\eta$ and $\zeta$ are the compositions of isomorphisms (1.2) of Theorem 1.1 with the explicit isomorphism (4.14).*

It is very important that we construct the isomorphisms of Theorems 1.1 and 1.2 explicitly, by constructing them as explicits maps of the standard Hochschild complexes. For any graded vector space $E$, the symmetric algebra $S^*(E \otimes t\Bbbk[t])$ carries a number of natural structures: of a Hopf algebra, of the free $\lambda$-ring generated by $E$, and of the Fock space for the Heisenberg algebra $H_E$ of $E$. Isomorphisms (1.3) allow us to transfer these structures to the total Hochschild homology $\bigoplus_{n\geq 0} \mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A})$ and compute explicitly the result.

In particular, one immediate application of our Theorem 1.2 is that $\bigoplus_{n\geq 0}\mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A})$ is the Fock space for the Heisenberg algebra $H_{\mathrm{HH}_\bullet(\mathcal{A})}$. This generalises the K-theoretic results of Segal [Seg96] and Wang [Wan00] inspired by the famous Heisenberg action of Nakajima [Nak97] and Grojnowski [Gro95]. In [GL25] Gyenge and the third author use the results of this paper to compute explicitly the induced Heisenberg action on $\bigoplus_{n\geq 0}\mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A})$ and prove it to be the same as the decategorification of the categorical Heisenberg action of [GKL21].

In this paper, we use the isomorphisms of Theorem 1.2 to compute the induced Hopf algebra and $\lambda$-ring structures on $\bigoplus_{n\geq 0}\mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A})$. Our third main result is:

**Theorem 1.3** (see Theorem 5.1)**.** *Let $\mathcal{A}$ be a small DG category. The isomorphisms of Theorem 1.2 identify the standard Hopf algebra operations $\mu$, $\Delta$, $u$, $\epsilon$, and $H$ on $S^*(\mathrm{HH}_\bullet(\mathcal{A})\otimes t\Bbbk[t])$ with the following operations on $\bigoplus_{n\geq 0}\mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A})$:*

- *The multiplication map $\mu$ given by the sum over all $n,m\geq 0$ of the maps*

$$\mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A})\otimes\mathrm{HC}_\bullet(\mathcal{S}^m\mathcal{A})\xrightarrow{\simeq}\mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A}\otimes\mathcal{S}^m\mathcal{A})\xrightarrow{\mathrm{Ind}_{S_n\times S_m}^{S_{n+m}}}\mathrm{HH}_\bullet(\mathcal{S}^{n+m}\mathcal{A}).$$

- *The comultiplication map $\Delta$ given by the sum over all $n\geq 0$ and all subsets $I\subseteq\{1,\dots,n\}$ of the maps*

$$\mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A})\xrightarrow{\mathrm{Res}_{S_I\times S_{\bar I}}^{S_n}}\mathrm{HH}_\bullet(\mathcal{S}^{|I|}\mathcal{A}\otimes\mathcal{S}^{n-|I|}\mathcal{A})\xrightarrow{\simeq}\mathrm{HH}_\bullet(\mathcal{S}^{|I|}\mathcal{A})\otimes\mathrm{HC}_\bullet(\mathcal{S}^{n-|I|}\mathcal{A}),$$

  *where $\bar I=\{1,\dots,n\}\setminus I$, where $S_I,S_{\bar I}<S_n$ are the subgroups that only permute the elements of $I$ and of $\bar I$, and where $|I|$ is the size of $I$.*
- *The unit map $u'$ and the counit map $\epsilon'$ given by the inclusion of and the projection onto $\mathrm{HH}_\bullet(\mathcal{S}^0\mathcal{A})\simeq\Bbbk$,*
- *The antipode map $H'$ given on $\mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A})$ by $(-1)^n\,\mathrm{Id}$.*

Finally, in Section 5.2 we compute the induced $\lambda$-ring structure on $\bigoplus_{n\geq 0}\mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A})$ and discuss the parallels with the $K$-theoretic results of Wang [Wan00].

When finishing our work on this paper, it was brought to our attention that the decompositions of Theorems 1.1 and 1.2 were independently and simultaneously established by Nordström in [Nor25a]. He establishes them via general considerations which do not yield explicit maps on the level of Hochschild complexes. It is not possible, using his approach, to compute the induced Fock space, Hopf algebra, and $\lambda$-ring structures on $\bigoplus_{n\geq 0}\mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A})$.

This paper is organized as follows. In Section 2 we give a quick overview of the standard tools for DG categories: tensor products, modules, Hochschild complexes, semidirect products by a finite group and Eilenberg-Zilber theorem for Hochschild complexes. In particular, for any functor $F:\mathcal{A}\to\mathcal{A}$ we have the twist ${}_F\mathcal{A}$ of the $\mathcal{A}$-$\mathcal{A}$ bimodule $\mathcal{A}$ and the Hochschild complex $\mathrm{HC}_\bullet(\mathcal{A},F)$ computing Hochschild homology of this bimodule. The emphasis is on explicit maps and constructions. In Section 3 we consider the case of the cycle $t_n=(1,2,\dots,n)$ acting on $\mathcal{A}^{\otimes n}$ and construct an explicit homotopy equivalence between the complex of coinvariants $\mathrm{HC}_\bullet(\mathcal{A}^{\otimes n};t_n)_{t_n}$ and $\mathrm{HC}_\bullet(\mathcal{A})$. In Section 4 we use this computation and Eilenberg-Zilber theorem to write down the explicit isomorphisms of Theorems 1.1 and 1.2. In Section 5 we use these explicit isomorphisms to compute to compute the Hopf algebra and $\lambda$-ring structures on the direct sum $\bigoplus_{n\geq 0}\mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A})$ in terms of standard algebraic constructions involving $\mathrm{HH}_\bullet(\mathcal{A})$.

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## 2. Preliminaries

2.1. **DG categories, modules and tensor products.** We briefly recall here the basics on DG categories (always assumed small in this paper) mostly to fix the notation, sending the reader to [Kel06] for a more detailed treatment. A DG category over a field $\Bbbk$ is a category $\mathcal{A}$ in which for any pair of objects $a, b$ the set ${}_a\mathcal{A}_b$ of morphisms $b \to a$ is a differential graded $\Bbbk$-module, i.e. a complex of vector spaces over $\Bbbk$. The inversion of order in the notation ${}_a\mathcal{A}_b$ for morphisms makes it easier to identify composable morphisms when working with Hochschild complexes as compositions of morphisms are described by morphisms of complexes ${}_a\mathcal{A}_b \otimes_\Bbbk {}_b\mathcal{A}_c \to {}_a\mathcal{A}_c$, $\alpha \otimes \beta \mapsto \alpha\beta$.

DG functors between DG categories are usual functors that are compatible with differentials on morphisms.

A *left DG module* $M$ over a DG category $\mathcal{A}$ is a DG functor $M : \mathcal{A} \to C_{dg}(\Bbbk)$ taking values in the DG category of complexes over $\Bbbk$. Similarly, a *right DG module* is a functor $\mathcal{A}^{op}$ from the opposite category. Hence for each object $a \in obj(\mathcal{A})$ we have a complex ${}_aM := M(a)$ with morphisms of complexes ${}_a\mathcal{A}_b \otimes_\Bbbk {}_bM \to {}_aM$ for any pair of objects $a, b$, and similarly for right DG modules.

A *tensor product* $\mathcal{A}_1 \otimes \mathcal{A}_2$ of two DG categories $\mathcal{A}_1$, $\mathcal{A}_2$ has the set of objects $\mathrm{Ob}(\mathcal{A}_1) \times \mathrm{Ob}(\mathcal{A}_2)$ with the morphisms

$$ {}_{a_1,a_2}(\mathcal{A}_1 \otimes \mathcal{A}_2)_{b_1,b_2} := {}_{a_1}(\mathcal{A}_1)_{b_1} \otimes_\Bbbk {}_{a_2}(\mathcal{A}_2)_{b_2} $$

and natural compositions (involving the Koszul sign rule) and units.

A *DG $\mathcal{A}_1$-$\mathcal{A}_2$-bimodule* $M$ if a left DG module over $\mathcal{A}_1 \otimes \mathcal{A}_2^{op}$ which is given by the complexes ${}_aM_b := M(a, b)$ similarly to the above. Central to this paper is the following special case of this notion. Let $F\colon \mathcal{A} \to \mathcal{A}$ be a DG functor. We define the bimodule

$$ {}_F\mathcal{A} \in \mathcal{A}\text{-}\mathbf{Mod}\text{-}\mathcal{A} $$

by setting

$$ {}_a({}_F\mathcal{A})_b := {}_{Fa}\mathcal{A}_b, $$

and letting $\mathcal{A}$ act naturally on the right and via $F$ on the left:

$$ \alpha.(\beta).\gamma = F(\alpha)\beta\gamma \in {}_{Fc}\mathcal{A}_d \qquad \forall\ \alpha \in {}_c\mathcal{A}_a,\ \beta \in {}_{Fa}\mathcal{A}_b,\ \gamma \in {}_b\mathcal{A}_d. $$

2.2. **Equivariant DG categories.** In this paper, we have to work with equivariant DG categories — DG categories equipped with *strong* group actions - and equivariant modules over them. To that end, a new approach was introduced in [GKL21, §4.8] and further refined in [GL25, §5.2]. It was designed for working with DG categories up to Morita equivalence and it differs e.g. from the one introduced by Kapranov and Ganter in [GK14]. We give below a small overview to set up notation.

Let $\mathcal{A}$ be a small DG category. A *strong* action of a finite group $G$ on $\mathcal{A}$ is an embedding of $G$ into the group of DG automorphisms of $\mathcal{A}$. In particular, $G$ acts by *automorphisms* rather than autoequivalences. It is a fairly rare situation but

in this paper we work with the action of $G = S_n$ and its subgroups, on the tensor powers $\mathcal{A}^{\otimes n}$. In this case the action is strong. See [Nor25b] for the discussion of Hochschild complexes in the setting of weaker actions of groups on DG categories.

**Definition 2.1** ([GKL21], Defn. 4.45)**.** The *semi-direct product* $\mathcal{A} \rtimes G$ is the following DG category:

- $\operatorname{Ob} \mathcal{A} \rtimes G = \operatorname{Ob} \mathcal{A}$,
- For any two objects $a, b \in \operatorname{Ob}(\mathcal{A} \rtimes G)$ their morphism complex is
$$ {}_a\mathcal{A} \rtimes G_b := \bigoplus_{g \in G} {}_{ga}\mathcal{A}_b.$$
We denote by $(\alpha, g)$ an element in the term corresponding to $g \in G$ and $\alpha \in {}_{ga}\mathcal{A}_b$. Then $\deg_{\mathcal{A}\rtimes G}(\alpha, g) = \deg_{\mathcal{A}} \alpha$ and $d_{\mathcal{A}\rtimes G}(\alpha, g) = (d_{\mathcal{A}}\alpha, g)$,
- The composition in $\mathcal{A} \rtimes G$ is given, similarly to the case of a twisted group ring, by the formula
$$(\alpha_1, g_1) \circ (\alpha_2, g_2) = (\alpha_1 \circ g_1.\alpha_2,\ g_1 g_2).$$
- For any $a \in \operatorname{Ob}(\mathcal{A} \rtimes G)$ the identity morphism of $a$ is $(\mathrm{Id}_a, 1_G)$.

With this definition, modules over $\mathcal{A} \rtimes G$ are $G$-equivariant modules over $\mathcal{A}$ in the sense of [GK14, §2.1.3].

**Lemma 2.2** ([GKL21], Lemma 4.41)**.** *There are mutually inverse isomorphisms of categories*
$$\mathbf{Mod}\text{-}(\mathcal{A} \rtimes G) \leftrightarrows \mathbf{Mod}^G\text{-}\mathcal{A},$$
$$h\mathcal{P}erf\text{-}(\mathcal{A} \rtimes G) \leftrightarrows h\mathcal{P}erf^G\text{-}\mathcal{A}.$$

The second part of the following lemma uses the notion of the perfect hull explained in Definition 4.22 or *loc.cit.*.

For any $g \in G$ the autoequivalence $g\colon \mathcal{A} \xrightarrow{\sim} \mathcal{A}$ extends to $\mathcal{A} \rtimes G$:

**Definition 2.3.** Let $\mathcal{A}$ be a small DG category and $G$ be a group acting strongly on $\mathcal{A}$. For any $g \in G$ define the autoequivalence
$$g\colon \mathcal{A} \rtimes G \xrightarrow{\sim} \mathcal{A} \rtimes G$$
to have the same action on objects as $g\colon \mathcal{A} \xrightarrow{\sim} \mathcal{A}$ and to act on morphisms by $g(\alpha, f) = (g(\alpha), gfg^{-1})$.

2.3. **Partitions.**

**Definition 2.4.** Let $n \geq 0$. An *(unordered) partition of* $n$ is an unordered collection $\underline{\lambda} := \{n_1, \ldots, n_m\}$ of strictly positive integers $n_i$ with $\sum_{i=1}^m n_i = n$. We denote this by $\underline{\lambda} \vdash n$.

The integers $n_i$ are the *parts* of $\underline{\lambda}$. We write $r(\underline{\lambda})$ for the *length* of $\underline{\lambda}$. By this we mean the total number of parts in $\underline{\lambda}$, i.e. $m$. We write $r_k(\underline{\lambda})$ for the number of parts of size $k$ in $\underline{\lambda}$, i.e. the number of $i$ such that $n_i = k$.

For $n = 0$, we use the convention that there exists a unique partition $\underline{0}$ of 0 with $r(\underline{0}) = r_k(\underline{0}) = 0$.

**Definition 2.5.** Let $n \geq 0$. An *ordered partition of* $n$ is $(n_1, \ldots, n_m) \in \mathbb{Z}^m_{\geq 0}$ with $\sum_{i=1}^m n_i = n$. We denote this by $(n_1, \ldots, n_m) \vdash n$.

We write Forg for the forgetful map from all ordered partitions of $n$ to all unordered partitions of $n$ which forgets the ordering of the parts.

**Definition 2.6.** Let $n \geq 0$ and let $\underline{\lambda} \vdash n$ be an unordered partition. Define $\underline{r}(\underline{\lambda})$ to be the ordered partition $((r_1(\underline{\lambda}), \ldots, r_n(\underline{\lambda}))$ of $r(\underline{\lambda})$.

2.4. **Symmetric powers of DG categories.** For any small DG category $\mathcal{A}$ and any $n \geq 0$ we write $\mathcal{A}^{\otimes n}$ for the $n$-fold tensor product over $\Bbbk$, viewed as an object in the category $\mathcal{A}$-$\mathcal{A}$-bimodules, cf. [Kel94, §6.1]. In other words

$$ {}_a\left(\mathcal{A}^{\otimes n}\right)_b := \bigoplus_{a_1,\ldots,a_{n-1} \in \mathcal{A}} {}_a\mathcal{A}_{a_1} \otimes_\Bbbk {}_{a_1}\mathcal{A}_{a_2} \otimes_\Bbbk \cdots \otimes {}_{a_{n-1}}\mathcal{A}_b. $$

For $n = 0$, we use the convention that $\mathcal{A}^{\otimes 0} = \Bbbk$. Our main object of study in this paper is introduced in the following definition.

**Definition 2.7.** Let $\mathcal{A}$ be a small DG category. Let $n \geq 0$ and let $(n_1, \ldots, n_m) \vdash n$ be an ordered partition. Define

$$ S_{n_1,\ldots,n_m} := S_{n_1} \times \cdots \times S_{n_m} < S_n, $$

and define $(n_1, \ldots, n_m)$*-th symmetric power of* $\mathcal{A}$ to be

$$ \mathcal{S}^{n_1,\ldots,n_m}\mathcal{A} := \mathcal{A}^{\otimes n} \rtimes \left(S_{n_1} \times \cdots \times S_{n_m}\right). $$

**Definition 2.8.** Let $\mathcal{A}$ be a small DG category. Let $n \geq 0$ and let $\underline{\lambda} \vdash n$ be a partition. Define $S_{\underline{\lambda}}$ and $\mathcal{S}^{\underline{\lambda}}\mathcal{A}$ to be the isomorphism classes of $S_{n_1,\ldots,n_m}$ and $\mathcal{S}^{n_1,\ldots,n_m}\mathcal{A}$ for any $(n_1, \ldots, n_m)$ such that $\mathrm{Forg}(n_1, \ldots, n_m) = \underline{\lambda}$.

We make similar definitions for graded vector spaces, complexes of vector spaces, etc. All of these can be viewed as a special case of the following definition, which we need to work with Hochschild complexes:

**Definition 2.9.** Let $E \in \text{Pre-Tr}^-(\mathbf{Mod}\text{-}\Bbbk)$ be a bounded above twisted complex over $\mathbf{Mod}$-$\Bbbk$, cf. [AL24, §3]. Let $n \geq 0$ and let $(n_1, \ldots, n_m) \vdash n$ be an ordered partition. Define $(n_1, \ldots, n_m)$*-th symmetric power of* $E$ to be

$$ \mathcal{S}^{n_1,\ldots,n_m}E := (E^{\otimes n})_{S_{n_1,\ldots,n_m}} \in \text{Pre-Tr}^-(\mathbf{Mod}\text{-}\Bbbk), $$

where the index denotes taking the coinvariants under the action of $S_{n_1,\ldots,n_m}$ which permutes the corresponding factors.

For any unordered partition $\underline{\lambda} \vdash n$ define $\mathcal{S}^{\underline{\lambda}}E$ to be the isomorphism class of $\mathcal{S}^{n_1,\ldots,n_m}E$ for any $(n_1, \ldots, n_m)$ such that $\mathrm{Forg}(n_1, \ldots, n_m) = \underline{\lambda}$.

2.5. **Hochschild homology of a bimodule.** Once we have the notion of a bimodule $M$ over a category $\mathcal{A}$, the usual formalism of resolutions and derived functors can be extended to the categorical level in a reasonably straightforward way, see [Kel94, §2 and §3], and the usual definition makes sense, cf. [Kel20, §6.8] for the case $M = \mathcal{A}$:

**Definition 2.10.** Let $\mathcal{A}$ be a small DG-category and let $M \in \mathcal{A}\text{-}\mathbf{Mod}\text{-}\mathcal{A}$. The *Hochschild homology of* $\mathcal{A}$ *with coefficients in* $M$ is

$$ \mathrm{HH}_\bullet(\mathcal{A}; M) := H^\bullet(M \overset{\mathbf{L}}{\otimes}_{\mathcal{A}\text{-}\mathcal{A}} \mathcal{A}), \tag{2.1} $$

See also [Lod97, §1.1.3] [Bar03, §3, Step 4][Nor25b, §3.2]. Here $\overset{\mathbf{L}}{\otimes}_{\mathcal{A}\text{-}\mathcal{A}}$ is the derived functor of the DG functor

$$ \otimes_{\mathcal{A}\text{-}\mathcal{A}} \colon \mathcal{A}\text{-}\mathbf{Mod}\text{-}\mathcal{A} \otimes_k \mathcal{A}\text{-}\mathbf{Mod}\text{-}\mathcal{A} \to \mathbf{Mod}\text{-}\Bbbk \tag{2.2} $$

where we tensor the left $\mathcal{A}$-action with the right $\mathcal{A}$-action and vice versa. More precisely, $\mathcal{A}$-$\mathcal{A}$-bimodules are, equivalently, right $\mathcal{A}^{\mathrm{opp}} \otimes_{\Bbbk} \mathcal{A}$-modules or left $\mathcal{A} \otimes_{\Bbbk} \mathcal{A}^{\mathrm{opp}}$-modules . We can view them as both left and right $\mathcal{A}^{\mathrm{opp}} \otimes_{\Bbbk} A$-modules via the canonical isomorphism on objects

$$\begin{aligned} \mathcal{A} \otimes_{\Bbbk} \mathcal{A}^{\mathrm{opp}} &\xrightarrow{\sim} \mathcal{A}^{\mathrm{opp}} \otimes_{\Bbbk} \mathcal{A}, \\ a \otimes b \quad &\mapsto \quad b \otimes a \end{aligned}$$

while for morphisms on the right hand side there is a factor $(\pm 1)$ determined by the Koszul rule. Then (2.2) is the functor of tensoring over this module structure. Explicitly, it sends any pair $E, F \in \mathcal{A}\text{-}\mathbf{Mod}\text{-}\mathcal{A}$ to the complex of $\Bbbk$-modules

$$E \otimes_{\mathcal{A}\text{-}\mathcal{A}} F := \Big( \bigoplus_{a,b \in obj(\mathcal{A})} {}_aE_b \otimes_{\Bbbk} {}_bF_a \Big) / \{ e \otimes \alpha.f.\beta - \beta.e.\alpha \otimes f \} \tag{2.3}$$

for all $e \in {}_aE_b$, $\alpha \in {}_b\mathcal{A}_c$, $f \in {}_cF_d$, $\beta \in {}_d\mathcal{A}_a$ and quadriples of objects $a, b, c, d$ in $\mathcal{A}$. Note that the term $e \otimes \alpha.f.\beta$ is in ${}_aE_b \otimes_{\Bbbk} {}_bF_a$ while $\beta.e.\alpha \otimes f$ is in ${}_dE_c \otimes_{\Bbbk} {}_cF_d$.

We can compute $\overset{\mathbf{L}}{\otimes}_{\mathcal{A}\text{-}\mathcal{A}}$ by taking an h-projective resolution in either variable. Using the standard bar-complex resolution $\bar{\mathcal{A}}$ of the diagonal bimodule $\mathcal{A}$, see e.g. [AL21, Section 2.11], we see that $\mathrm{HH}_\bullet(\mathcal{A}; M)$ are isomorphic to the cohomology of the convolution of the *Hochschild complex* $\mathrm{HC}_\bullet(\mathcal{A}; M)$ *with coefficients in* $M$:

$$\cdots \to \bigoplus_{a,b,c \in \mathcal{A}} {}_aM_c \otimes_{\Bbbk} {}_c\mathcal{A}_b \otimes_{\Bbbk} {}_b\mathcal{A}_a \to \bigoplus_{a,b \in \mathcal{A}} {}_aM_b \otimes_{\Bbbk} {}_b\mathcal{A}_a \to \bigoplus_{a \in \mathcal{A}} {}_aM_a \tag{2.4}$$

with the differentials defined by

$$\begin{aligned} m_0 \otimes \alpha_1 \otimes \cdots \otimes \alpha_n \quad \mapsto \quad & m_0.\alpha_1 \otimes \cdots \otimes \alpha_{n-1} + \\ & + \sum_{i=1}^{n-1} (-1)^i \, m_0 \otimes \cdots \otimes \alpha_i\alpha_{i+1} \otimes \cdots \otimes \alpha_n + \\ & + (-1)^{n+|\alpha_n|(|m_0|+\cdots+|\alpha_{n-1}|)} \alpha_n.m_0 \otimes \alpha_1 \otimes \cdots \otimes \alpha_{n-1} \end{aligned}$$

**Main Example.** For $M = \mathcal{A}$ we get Hochschild homology $\mathrm{HH}_\bullet(\mathcal{A}; \mathcal{A})$. If $F : \mathcal{A} \to \mathcal{A}$ is a functor then we define the *$F$-twisted Hochschild homology* $\mathrm{HH}_\bullet(\mathcal{A}; F)$ to be the Hochschild homology $\mathrm{HH}_\bullet(\mathcal{A}; {}_F\mathcal{A})$ of $\mathcal{A}$ with coefficients in the bimodule ${}_F\mathcal{A}$ induced by $F$.

The main computation of this paper is carried out in the case when $F$ is the permutation functor $\mathcal{A}^{\otimes n} \to \mathcal{A}^{\otimes n}$ given by the cycle $t_n = (1, 2, \ldots n)$.

2.6. **Shuffles.** The usual Künneth isomorphism works for DG categories with very little change. As in the case of $\Bbbk$-algebras, one starts by defining shuffles:

**Definition 2.11.** Let $p, q \in \mathbb{Z}_{\leq 0}$. Define the subset of $(p, q)$-shuffles

$$S_{p,q} := \left\{ \sigma \in S_{p+q} \;\middle|\; \sigma(i) < \sigma(j) \text{ if } \begin{array}{c} 1 \leq i < j \leq p, \\ \text{or} \\ p+1 \leq i < j \leq p+q. \end{array} \right\} \tag{2.5}$$

The sign $(-1)^\sigma$ of a shuffle $\sigma \in S_{p,q}$ is its sign as a permutation in $S_n$.

In other words, $(p, q)$-shuffles are the elements of the permutation group $S_{p+q}$ which preserve relative order of the first $p$ and the last $q$ elements. Later in the paper we will need a computation of the number defined as follows:

**Definition 2.12.** Let $p, q \in \mathbb{Z}_{\leq 0}$. Define

$$Y_{p,q} := \begin{cases} \binom{\lfloor \frac{p+q}{2} \rfloor}{\lfloor \frac{p}{2} \rfloor} & \text{either } p \text{ or } q \text{ is even,} \\ 0, & \text{otherwise.} \end{cases}$$

**Lemma 2.13.** *Let $p, q \in \mathbb{Z}_{\leq 0}$. Then*

$$Y_{p,q} = \sum_{\sigma \in S_{p,q}} (-1)^\sigma. \tag{2.6}$$

*Proof.* We prove by induction on $p+q$. All $(p,q)$-shuffles $\sigma$ can be divided into those where the $\sigma^{-1}(1) = 1$ and $\sigma^{-1}(1) = p+1$. The former correspond bijectively to $(p-1,q)$-shuffles of the same parity, the latter correspond bijectively to $(p,q-1)$-shuffles whose parity differs by $(-1)^p$. Thus

$$\sum_{\sigma \in S_{p,q}} (-1)^\sigma = \sum_{\sigma \in S_{p-1,q}} (-1)^\sigma + (-1)^p \sum_{\sigma \in S_{p,q-1}} (-1)^\sigma.$$

By direct computation, we also have

$$Y_{p,q} = Y_{p-1,q} + (-1)^p Y_{p,q-1}. \tag{2.7}$$

It remains to establish (2.6) in the cases where either $p = 0$ or $q = 0$, which is trivial. □

We will also need below a slight generalisation of shuffles:

**Definition 2.14.** Let $P, Q \subset \{1, \dots, n\}$ be two disjoint subsets. Define the subset of $(P,Q)$-shuffles

$$S_{P,Q} := \left\{ \sigma \in S_n \;\middle|\; \begin{array}{lll} \sigma(i) < \sigma(j) & \text{if} & i < j \text{ with } i,j \in P \text{ or } i,j \in Q, \\ \sigma(i) = i & \text{if} & i \neq P \cup Q \end{array} \right\} \tag{2.8}$$

The sign $(-1)^\sigma$ of a shuffle $\sigma \in S_{P,Q}$ is its sign as a permutation in $S_n$.

2.7. **Künneth isomorphism.** Let $\mathcal{A}$ and $\mathcal{B}$ be two DG categories. The well-known Künneth isomorphism

$$\mathrm{HH}_\bullet(\mathcal{A}; M) \otimes_{\Bbbk} \mathrm{HH}_\bullet(\mathcal{B}; N) \simeq \mathrm{HH}_\bullet(\mathcal{A} \otimes_{\Bbbk} \mathcal{B}; M \otimes_{\Bbbk} N) \tag{2.9}$$

can be written in terms of the Hochschild complexes (2.4) as follows. First we define it on the level of bar complexes:

**Definition 2.15.** Let $\mathcal{A}$ and $\mathcal{B}$ be two DG categories. Let $\bar{\mathcal{A}}$ and $\bar{\mathcal{B}}$ be their bar-complexes viewed as twisted complexes of $\mathcal{A}$-$\mathcal{A}$- and $\mathcal{B}$-$\mathcal{B}$-bimodules respectively [AL21, Defn. 2.24]. Similarly, let $\overline{\mathcal{A} \otimes_{\Bbbk} \mathcal{B}}$ be the bar complex of the tensor product category $\mathcal{A} \otimes_{\Bbbk} \mathcal{B}$.

Define the Eilenberg-Zilber map

$$\mathrm{EZ} \colon \bar{\mathcal{A}} \otimes_{\Bbbk} \bar{\mathcal{B}} \to \overline{\mathcal{A} \otimes_{\Bbbk} \mathcal{B}}$$

of twisted complexes of $\mathcal{A} \otimes_{\Bbbk} \mathcal{B}$-$\mathcal{A} \otimes_{\Bbbk} \mathcal{B}$-bimodules to comprise the components

$$\mathcal{A}^{\otimes n} \otimes_{\Bbbk} \mathcal{B}^{\otimes m} \xrightarrow{\sum_{\sigma \in S_{n-2,m-2}} (-1)^\sigma \iota_\sigma} \mathcal{A}^{\otimes n+m-2} \otimes_{\Bbbk} \mathcal{B}^{\otimes n+m-2} \xrightarrow{\sim} (\mathcal{A} \otimes_{\Bbbk} \mathcal{B})^{\otimes n+m-2} \tag{2.10}$$

where $n, m \geq 2$ and for any $(n-2, m-2)$-shuffle $\sigma \in S_{n-2,m-2}$ the map

$$\iota_\sigma \colon \mathcal{A}^{\otimes n} \otimes_{\Bbbk} \mathcal{B}^{\otimes m} \to \mathcal{A}^{\otimes n+m-2} \otimes_{\Bbbk} \mathcal{B}^{\otimes n+m-2}$$

is given first by the repeated application of the identity insertion map

$$\begin{aligned} \mathcal{A}^{\otimes 2} &\to \mathcal{A}^{\otimes 3} \\ \alpha_1 \otimes \alpha_2 &\mapsto \alpha_1 \otimes \mathrm{Id}_{a_1} \otimes \alpha_2 \qquad \forall\, \alpha_i \in \mathrm{Hom}_{\mathcal{A}}(a_i, a_{i-1}), \end{aligned}$$

to insert $m-2$ identity maps into $\mathcal{A}^{\otimes n}$ onto the positions $\sigma((n-2)+1)+1$, $\sigma((n-2)+2)+1$, ..., $\sigma((n-1)+m-2)+1$ in $\mathcal{A}^{\otimes(n+m-2)}$, and then inserting $n-2$ identity maps onto the positions $\sigma(1)+1, \ldots, \sigma(n-2)+1$ in $\mathcal{B}^{\otimes(n+m-2)}$.

Since for any DG category $\mathcal{C}$ and a bimodule $L$ we have $\mathrm{HC}_\bullet(\mathcal{C}; M) \simeq \bar{\mathcal{C}} \otimes_{\mathcal{C}\text{-}\mathcal{C}} L$, we can now define the Eilenberg-Zilber morphism for the Hochschild complexes:

**Definition 2.16.** Let $\mathcal{A}$ and $\mathcal{B}$ be two DG categories. The Eilenberg-Zilber morphism

$$\mathrm{EZ}\colon\ \mathrm{HC}_\bullet(\mathcal{A}; M) \otimes_{\Bbbk} \mathrm{HC}_\bullet(\mathcal{B}; N) \to \mathrm{HC}_\bullet(\mathcal{A} \otimes \mathcal{B}; M \otimes_{\Bbbk} N), \tag{2.11}$$

is the composition

$$\begin{array}{c} \left(\bar{\mathcal{A}} \otimes_{\mathcal{A}\text{-}\mathcal{A}} M\right) \otimes_{\Bbbk} \left(\bar{\mathcal{B}} \otimes_{\mathcal{B}\text{-}\mathcal{B}} N\right) \\ \downarrow \sim \\ \left(\bar{\mathcal{A}} \otimes_{\Bbbk} \bar{\mathcal{B}}\right) \otimes_{\mathcal{A}\otimes_{\Bbbk}\mathcal{B}\text{-}\mathcal{A}\otimes_{\Bbbk}\mathcal{B}} (M \otimes_{\Bbbk} N) \\ \downarrow (2.10) \\ \overline{\mathcal{A} \otimes_{\Bbbk} \mathcal{B}} \otimes_{\mathcal{A}\otimes_{\Bbbk}\mathcal{B}\text{-}\mathcal{A}\otimes_{\Bbbk}\mathcal{B}} (M \otimes_{\Bbbk} N)\,. \end{array} \tag{2.12}$$

Explicitly, for any Hochschild chains

$$\underline{\alpha} = m \otimes \cdots \otimes \alpha_n \in \mathrm{HC}_n(\mathcal{A}; M)$$

$$\underline{\beta} = n \otimes \cdots \otimes \beta_m \in \mathrm{HC}_m(\mathcal{B}; N)$$

we have

$$\mathrm{EZ}\left(\underline{\alpha} \otimes \underline{\beta}\right) = \sum_{\sigma \in S_{n,m}} (-1)^{\sigma} (-1)^{\deg_\sigma(\underline{\alpha}, \underline{\beta})} (m \otimes n) \otimes \cdots \otimes (\alpha_i \otimes \mathrm{Id}) \otimes \ldots (\mathrm{Id} \otimes \beta_j) \otimes \ldots \tag{2.13}$$

where in the summand indexed by $\sigma \in S_{n,m}$ the factors $(\alpha_i \otimes \mathrm{Id})$ and $(\mathrm{Id} \otimes \beta_j)$ occur in the positions $\sigma(i)$ and $\sigma(n+j)$, respectively. The second sign is computed by setting $d_\sigma(\underline{\alpha}, \underline{\beta})$ to be the sum of $\deg(\alpha_i)\deg(\beta_j)$ for all $0 \le i \le n$ and $0 \le j \le m$ such that the factor containing $\alpha_i$ occurs to the right of the factor containing $\beta_j$.

As in [Lod97, §4.2] the morphism EZ induces isomorphism of the Hochschild homology groups. The inverse isomorphism is induced by the Alexander-Whitney map [ML63, Corollary X.7.2]

$$\mathrm{AW}((a_0 \otimes b_0) \otimes (a_1 \otimes b_1) \otimes \ldots \otimes (a_n \otimes b_n)) =$$

$$\sum_{p=0}^{n} (-1)^{(2.14)} \left(a_0 a_1 a_2 \ldots a_p \otimes a_{p+1} \otimes \ldots \otimes a_n\right) \otimes \left(b_{p+1} \ldots b_n b_0 \otimes b_1 \otimes \ldots \otimes b_p\right)$$

where the sign is given by

$$p(n-p) + \sum_{i=0}^{n-1} \deg b_i \left(\sum_{j=i+1}^{n} \deg a_j\right) + \left(\sum_{i=p+1}^{n} \deg b_i\right)\left(\sum_{j=0}^{p} \deg b_j\right). \tag{2.14}$$

2.8. **Shuffle product with the constants.** We need to introduce the following notion:

**Definition 2.17.** Let $\mathcal{A}$ be a DG category. Let

$$\underline{\alpha} = \alpha_0 \otimes \alpha_1 \otimes \cdots \otimes \alpha_n \in \mathrm{HC}_n(\mathcal{A})$$

be a Hochschild $(n+1)$-chain in $\mathcal{A}$ and let

$$\underline{\kappa} = \kappa_0 \otimes \kappa_1 \otimes \cdots \otimes \kappa_m \in \mathrm{HC}_m(\Bbbk)$$

be a Hochschild $(m+1)$-chain in $\Bbbk$. Define the shuffle

$$\mathrm{Sh}\,(\underline{\alpha}, \underline{\kappa}) \in \mathrm{HC}_{n+m}(\mathcal{A}) \tag{2.15}$$

to be the image of $\underline{\alpha} \otimes \underline{\kappa}$ under the composition

$$\mathrm{HC}_\bullet(\mathcal{A}) \otimes \mathrm{HC}_\bullet(\Bbbk) \xrightarrow{K} \mathrm{HC}_\bullet(\mathcal{A} \otimes_\Bbbk \Bbbk) \xrightarrow{\sim} \mathrm{HC}_\bullet(\mathcal{A})$$

where the first map is the Künneth morphism (2.11) and the second map is induced by the canonical DG category isomorphism $\mathcal{A} \otimes_\Bbbk \Bbbk \xrightarrow{\sim} \mathcal{A}$.

Our principal application is the case $\underline{\kappa} = 1^{\otimes(m+1)}$. Explicitly, $\mathrm{Sh}\left(\underline{\alpha}, 1^{\otimes(m+1)}\right)$ is the sum over all $\sigma \in S_{n,m}$ of the $n+m$-chains in $\mathcal{A}$ obtained by using $\sigma$ to shuffle $m$ appropriate identity morphisms into $\underline{\alpha}$. The sign of each summand is $(-1)^\sigma$.

## 3. Long cycle case

Let $\mathcal{A}$ be a small DG category and let $n \geq 1$. The permutation group $S_n$ acts on the tensor power $\mathcal{A}^{\otimes n}$ by permuting the factors. For example, the long cycle $t_n := (1 \ldots n) \in S_n$ acts by the isomorphism of categories

$$t_n \colon \mathcal{A}^{\otimes n} \to \mathcal{A}^{\otimes n}$$

which maps

$$a_1 \otimes a_2 \otimes \cdots \otimes a_n \to a_n \otimes a_1 \otimes \ldots a_{n-1}$$

on objects and correspondingly on the morphisms. We use the same notation $t_n$ for the above permutation functor and write just $t$ for $t_n$ when no confusion is possible. Since the cycle $t_n$ commutes with itself, the isomorphism functor induced by it induces an isomorphism

$$t_n \colon \mathrm{HC}_\bullet(\mathcal{A}^{\otimes n}; t) \xrightarrow{\sim} \mathrm{HC}_\bullet(\mathcal{A}^{\otimes n}; t).$$

Let $\mathrm{HC}_\bullet(\mathcal{A}^{\otimes n}; t_n)_{t_n}$ denote the coinvariants of the action of $t$, that is, the vector space quotient by the subspace generated by $\underline{\alpha} - t_n.\underline{\alpha}$ for all $\underline{\alpha} \in \mathrm{HC}_\bullet(\mathcal{A}^{\otimes n}; t_n)$.

The main result of this section is the construction of two mutually inverse homotopy equivalences

$$\mathrm{HC}_\bullet(\mathcal{A}^{\otimes n}; t_n)_{t_n} \underset{g}{\overset{f}{\rightleftarrows}} \mathrm{HC}_\bullet(\mathcal{A}). \tag{3.1}$$

3.1. **Notation.** For a chain

$$\underline{\alpha}_0 \otimes \cdots \otimes \underline{\alpha}_{m-1} \in \mathrm{HC}_{m-1}(\mathcal{A}^n; t)$$

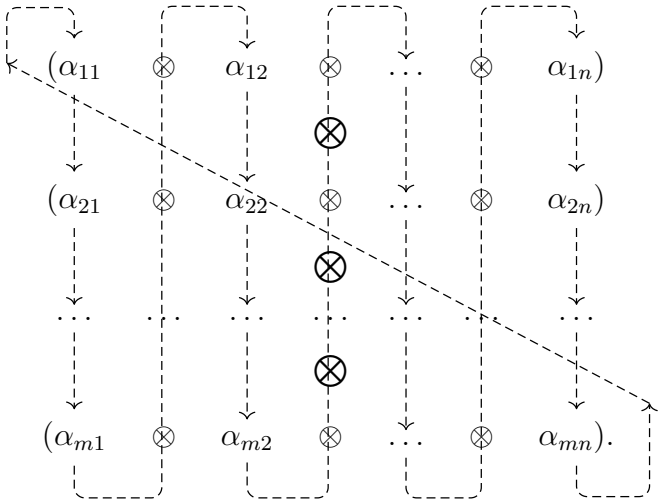
we will shift indexing and decompose it as a product of basic chains as follows

$$\underline{\alpha}_i = \alpha_{(i+1)1} \otimes \cdots \otimes \alpha_{(i+1)n} \tag{3.2}$$

for some

$$a_1, \cdots, a_{nm+1} \in \mathcal{A} \text{ with } a_1 = a_{nm+1},$$
$$\alpha_{ij} \in {}_{a_{(j-1)m+i+1}}\mathcal{A}_{a_{(j-1)m+i}}$$

The index shift in (3.2) is due to us wanting to use the matrix notation: the chain above can be visualised as




(3.3)

where the dotted line denotes the order in which the morphisms $\alpha_{ij}$ can be composed in. We denote such chain simply by the $m \times n$ matrix

$$\begin{pmatrix} \alpha_{11} & \alpha_{12} & \ldots & \alpha_{1n} \\ \alpha_{21} & \alpha_{22} & \ldots & \alpha_{2n} \\ & & \ldots & \\ \alpha_{m1} & \alpha_{m2} & \ldots & \alpha_{mn}. \end{pmatrix} \tag{3.4}$$

Note that for $n = 1$ we simply have

$$\mathrm{HC}_\bullet(\mathcal{A}^n; t) = \mathrm{HC}_\bullet(\mathcal{A}),$$

so the chain

$$\alpha_1 \otimes \ldots \alpha_m \in \mathrm{HC}_{m-1}(\mathcal{A})$$

is represented in the above notation by the column vector

$$\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \ldots \\ \alpha_m \end{pmatrix}. \tag{3.5}$$

Again, to facilitate the matrix notation we switched our notation for a chain in $\mathrm{HC}_{m-1}(\mathcal{A})$ from $\alpha_0 \otimes \cdots \otimes \alpha_{m-1}$ to $\alpha_1 \otimes \cdots \otimes \alpha_m$.

For example, we write a basic 4-chain in $\mathrm{HC}_3(\mathcal{A})$ as

$$\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{pmatrix}.$$

Each $\alpha_i$ is a morphism $a_{i+1} \to a_i$ for some $a_1, \dots, a_5 \in \mathcal{A}$ with $a_5 = a_1$:

$$a_1 \xrightarrow{\alpha_4} a_4 \xrightarrow{\alpha_3} a_3 \xrightarrow{\alpha_2} a_2 \xrightarrow{\alpha_1} a_1.$$

Similarly, we write a basic 3-chain in $\mathrm{HC}_2(\mathcal{A}^2; t)$ as

$$\begin{pmatrix} \alpha_{11} & \alpha_{12} \\ \alpha_{21} & \alpha_{22} \\ \alpha_{31} & \alpha_{32} \end{pmatrix}.$$

Each $\alpha_{ij}$ is a morphism $a_{(i+1)j} \to a_{ij}$ for some $a_{11}, a_{12}, \dots, a_{41}, a_{42} \in \mathcal{A}$ with $a_{41} = a_{12}$ and $a_{42} = a_{11}$:

$$a_{11} \xrightarrow{\alpha_{32}} a_{32} \xrightarrow{\alpha_{22}} a_{22} \xrightarrow{\alpha_{12}} a_{12} \xrightarrow{\alpha_{31}} a_{31} \xrightarrow{\alpha_{21}} a_{21} \xrightarrow{\alpha_{11}} a_{11}.$$

### 3.2. Map $f$.

**Definition 3.1.** Let $\mathcal{A}$ be a small DG category. For each $n \geq 1$ define the map

$$f\colon \mathrm{HC}_\bullet(\mathcal{A}^{\otimes n}; t)_t \to \mathrm{HC}_\bullet(\mathcal{A})$$

by setting for each $m \geq 1$ the map $f\colon \mathrm{HC}_{m-1}(\mathcal{A}^{\otimes n}; t)_t \to \mathrm{HC}_{m-1}(\mathcal{A})$ to be

$$\begin{pmatrix} \alpha_{11} & \alpha_{12} & \dots & \alpha_{1n} \\ \alpha_{21} & \alpha_{22} & \dots & \alpha_{2n} \\ & & \dots & \\ \alpha_{m1} & \alpha_{m2} & \dots & \alpha_{mn}. \end{pmatrix} \quad \mapsto \quad \frac{1}{n} \sum_{s=1}^{n} \begin{pmatrix} \alpha_{1(s+1)} \dots \alpha_{ms-1} \alpha_{1s} \\ \alpha_{2s} \\ \dots \\ \alpha_{ms} \end{pmatrix} \tag{3.6}$$

### 3.3. Map $g$.

**Definition 3.2.** Let $\mathcal{A}$ be a small DG category. For each $n \geq 1$ define the map

$$g\colon \mathrm{HC}_\bullet(\mathcal{A}) \to \mathrm{HC}_\bullet(\mathcal{A}^{\otimes n}; t)_t$$

by setting for each $m \geq 1$ the map $g\colon \mathrm{HC}_{m-1}(\mathcal{A}) \to \mathrm{HC}_{m-1}(\mathcal{A}^{\otimes n}; t)_t$ to be

$$\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \dots \\ \alpha_m \end{pmatrix} \quad \mapsto \quad \sum_{c \in \{1,\dots,n\}^m \text{ with } c_1 = 1} (-1)^{\sigma_c} \begin{pmatrix} \beta_{11} & \beta_{12} & \dots & \beta_{1n} \\ \beta_{21} & \beta_{22} & \dots & \beta_{2n} \\ & & \dots & \\ \beta_{m1} & \beta_{m2} & \dots & \beta_{mn}. \end{pmatrix} \tag{3.7}$$

Here:

- The summation is taken over all $c = (c_1, \dots, c_m)$ with $c_i \in \{1, \dots, n\}$ and $c_1 = 1$. We think of each $c_i$ as a choice of a column position in the $i$-th row of the matrix $(\beta_{ij})$ and we always choose the first position in the first row.
- We define $\sigma_c \in S_m$ to be the permutation where $\sigma_c(i)$ is the number of $c_k$ with $c_k < c_i$ or with $c_k = c_i$ and $k \leq i$.
- We define

$$\beta_{ij} := \begin{cases} \alpha_{\sigma_c(i)} & j = c_i, \\ \mathrm{Id} & j \neq c_i, \end{cases}$$

This definition can be understood as follows: take the matrix $(\beta_{ij})$, start at $\beta_{11}$, and follow the composable order of its entries (down each column and then to the top of the column to its right), placing $\alpha_1, \dots, \alpha_n$, in that order, in the chosen column position in each row. The remaining entries are filled with the identity maps. We obtain a matrix each whose row contains exactly one $\alpha_i$. The permutation $\sigma_c$ corresponds to the new ordering on $\alpha_i$'s given by counting from the top to the bottom row.

For example, for $n = 2$ and $m = 4$ the map $g$ is the map

$$\begin{pmatrix}\alpha_1\\ \alpha_2\\ \alpha_3\\ \alpha_4\end{pmatrix} \mapsto \begin{pmatrix}\alpha_1 & \mathrm{Id}\\ \alpha_2 & \mathrm{Id}\\ \alpha_3 & \mathrm{Id}\\ \alpha_4 & \mathrm{Id}\end{pmatrix} + \begin{pmatrix}\alpha_1 & \mathrm{Id}\\ \alpha_2 & \mathrm{Id}\\ \alpha_3 & \mathrm{Id}\\ \mathrm{Id} & \alpha_4\end{pmatrix} - \begin{pmatrix}\alpha_1 & \mathrm{Id}\\ \alpha_2 & \mathrm{Id}\\ \mathrm{Id} & \alpha_4\\ \alpha_3 & \mathrm{Id}\end{pmatrix} + \begin{pmatrix}\alpha_1 & \mathrm{Id}\\ \alpha_2 & \mathrm{Id}\\ \mathrm{Id} & \alpha_3\\ \mathrm{Id} & \alpha_4\end{pmatrix} + \begin{pmatrix}\alpha_1 & \mathrm{Id}\\ \mathrm{Id} & \alpha_4\\ \alpha_2 & \mathrm{Id}\\ \alpha_3 & \mathrm{Id}\end{pmatrix} - \begin{pmatrix}\alpha_1 & \mathrm{Id}\\ \mathrm{Id} & \alpha_3\\ \alpha_2 & \mathrm{Id}\\ \mathrm{Id} & \alpha_4\end{pmatrix} + \begin{pmatrix}\alpha_1 & \mathrm{Id}\\ \mathrm{Id} & \alpha_3\\ \mathrm{Id} & \alpha_4\\ \alpha_2 & \mathrm{Id}\end{pmatrix} + \begin{pmatrix}\alpha_1 & \mathrm{Id}\\ \mathrm{Id} & \alpha_2\\ \mathrm{Id} & \alpha_3\\ \mathrm{Id} & \alpha_4\end{pmatrix}$$

3.4. **Homotopy $\Phi$.** We first need the following auxilliary construction:

**Definition 3.3.** Let $m, n \geq 1$. We define the map

$$B_{p,q,d,r,\underline{c}}\colon \mathrm{HC}_{m-1}(\mathcal{A}^{\otimes n}) \to \mathrm{HC}_m(\mathcal{A}^{\otimes n}) \tag{3.8}$$

parametrised by

- $p \in \{1, \ldots, m\}$,
- $q \in \{2, \ldots, n\}$,
- $d \in \{1, \ldots, p\}$,
- $r \in \{0, \ldots, d-1\}$ if $q < n$ and $r = 0$ if $q = n$,
- $\underline{c} = (c_1, \ldots, c_r) \in \{1, \ldots, n-q\}^r$,

to be the map which sends any chain

$$\underline{\alpha} := \begin{pmatrix}\alpha_{11} & \alpha_{12} & \ldots & \alpha_{1n}\\ \alpha_{21} & \alpha_{22} & \ldots & \alpha_{2n}\\ & & \ldots & \\ \alpha_{m1} & \alpha_{m2} & \ldots & \alpha_{mn}.\end{pmatrix}$$

to the chain $\underline{\beta} = (\beta_{ij})$ defined by

| Row group: | $i$ | $j$ | $\beta_{ij}$ |
|---|---|---|---|
| 1 | 1 | 1<br>otherwise | $\alpha_{(p+1)q} \ldots \alpha_{mn}\alpha_{11}$<br>Id |
| 2 | $2, \ldots, r+1$ | $c_{i-1}$<br>otherwise | $\alpha_{\left(\sigma_{\underline{c}}(i-1)+1\right)1}$<br>Id |
| 3 | $r+2, \ldots, d$ | $n-q+1$<br>otherwise | $\alpha_{i1}$<br>Id |
| 4 | $d+1, \ldots, m+d-p$ | $n-q+2, \ldots, n$<br>otherwise | $\alpha_{(i+p-d)(j-n+q-1)}$<br>Id |
| 5 | $(m+d-p+1)$ | $n-q+2, \ldots, n$<br>otherwise | $\alpha_{1(j-n-q)} \ldots \alpha_{d(j-n-q)}$<br>Id |
| 6 | $m+d-p+2, \ldots, m+1$ | $n-q+1, \ldots, n$<br>otherwise | $\alpha_{(i-m+p-1)(j-n+q)}$<br>Id |

where $\sigma_{\underline{c}} \in S_r$, similar to Defn. 3.2, is the permutation of the rows which arises when we distribute $\alpha_{21}, \ldots, \alpha_{(r+1)1}$ in composable order onto the positions in the rows $2, \ldots, r+1$ chosen by $\sigma_{\underline{c}}$.

**Definition 3.4.** For each $n \geq 1$ define a degree $-1$ map

$$\Phi_n\colon \mathrm{HC}_\bullet(\mathcal{A}^{\otimes n}; t) \to \mathrm{HC}_\bullet(\mathcal{A}^{\otimes n}; t)$$

to be the map

$$\frac{1}{n}\sum_{s=1}^{n}\Phi'_n\circ t^{s-1}, \tag{3.9}$$

where $\Phi'_n$ is the map which sends any chain

$$\underline{\alpha} := \begin{pmatrix} \alpha_{11} & \alpha_{12} & \dots & \alpha_{1n} \\ \alpha_{21} & \alpha_{22} & \dots & \alpha_{2n} \\ & & \dots & \\ \alpha_{m1} & \alpha_{m2} & \dots & \alpha_{mn}. \end{pmatrix}$$

to

$$\sum_{p,q,d,r,\underline{c}}\sum_{\tau}\sum_{\upsilon}(-1)^{\sigma_{\underline{c}}}(-1)^{\tau}(-1)^{\upsilon}(-1)^{m+(m+d)(p+d)}\tau\upsilon B_{p,q,d,r,\underline{c}}\left(\underline{\alpha}\right). \tag{3.10}$$

where the summation ranges for $p, q, d, r, \underline{c}$ are as in Definition 3.3, $\tau$ ranges over $S_{\{r+2,\dots,m-p+d\},\{d+1,\dots,p\},m+1}$, $\upsilon$ ranges over $S_{\{2,\dots,r+1\},\{r+2,\dots,m+1\},m+1}$. The shuffles $\tau$ and $\upsilon$ act by permuting the rows of the matrix form of the chain. Finally, the permutation $\sigma_{\underline{c}} \in S_r$ is as in Definition 3.3.

**Theorem 3.1.** *We have an equality of maps* $\mathrm{HC}_\bullet(\mathcal{A}^{\otimes n};t)_t \to \mathrm{HC}_\bullet(\mathcal{A}^{\otimes n};t)_t$*:*

$$d\Phi_n = g_n f_n - \mathrm{Id}\,.$$

*Proof.* Let $f'_n$ be the $s=1$ summand of $f_n$, so that $f_n = \frac{1}{n}\sum_{s=1}^{n} f'_n\circ t^{s-1}$. It suffices to show that we have an equality of maps $\mathrm{HC}_\bullet(\mathcal{A}^{\otimes n};t) \to \mathrm{HC}_\bullet(\mathcal{A}^{\otimes n};t)$:

$$d\Phi'_n = g_n f'_n - \mathrm{Id}\,.$$

For any $m \geq 1$ take any basic Hochschild chain $\alpha \in \mathrm{HC}_{m-1}(\mathcal{A}^{\otimes n})$

$$\underline{\alpha} := \begin{pmatrix} \alpha_{11} & \alpha_{12} & \dots & \alpha_{1n} \\ \alpha_{21} & \alpha_{22} & \dots & \alpha_{2n} \\ & & \dots & \\ \alpha_{m1} & \alpha_{m2} & \dots & \alpha_{mn}. \end{pmatrix}$$

We need to show that

$$d\left(\Phi'_n\left(\underline{\alpha}\right)\right) = -\Phi'_n\left(d\underline{\alpha}\right) + g_n f'_n(\underline{\alpha}) - \underline{\alpha}. \tag{3.11}$$

We verify this by going through each summand in the sum (3.10) for $\Phi'_n(\underline{\alpha})$ on the LHS of (3.11) and differentiating it. Each summand of the result is a basic chain of $\mathrm{HC}_{m-1}(\mathcal{A}^{\otimes n})$ given by an $(m\times n)$-matrix involving $\alpha_{ij}$ with a coefficient $\pm 1$ coming from the the sign twists in (3.10) and in the Hochschild differential. For each such matrix, we point out that it occurs exactly one more time in the sum (3.10) for $d(\Phi'_n(\underline{\alpha}))$ on the LHS, in the sum (3.10) for $\Phi'_n(d\underline{\alpha})$ on the RHS, or in the term $g_n f'_n(\underline{\alpha}) - \underline{\alpha}$. We point out where it occurs and verify that the signs are such that the cancellation occurs.

At the end of the proof, we will list the special summands of $d(\Phi'_n(\underline{\alpha}))$ on the LHS which cancel out those in the the term $g_n f'_n(\underline{\alpha}) - \underline{\alpha}$. For most of the proof, we will take the summands of $d(\Phi'_n(\underline{\alpha}))$ on the LHS, and identify the corresponding summands of $d(\Phi'_n(\underline{\alpha}))$ on the LHS or of $\Phi'_n(d\underline{\alpha})$ on the RHS which cancel them out. Our analysis makes it clear that all of the summands of $d(\Phi'_n(\underline{\alpha}))$ on the RHS are cancelled out.

We first assume that $\tau = \upsilon = \mathrm{Id}_{S_{m+1}}$. This is the main substance of our analysis. We now consider the following cases:

$p, q, d, r$, and $\underline{c}$ generic:

We start with generic values of $p, q, d, r$, and $\underline{c}$ in (3.10) for which all six row ranges listed in the table defininning $B_{p,q,d,r\underline{c}}(\underline{\alpha})$ in Defn. 3.3 are present and non-empty. Explicitly, this means that $r \geq 2$, $d \geq r+1$, $d < p < m$, and $q < n$.

We produce the cancelling summand for each summand of $dB_{p,q,d,r\underline{c}}(\underline{\alpha})$. Denote by $d_i B_{p,q,d,r\underline{c}}(\underline{\alpha})$ the summand where $i$th and $(i+1)$st rows of $B_{p,q,d,r\underline{c}}(\underline{\alpha})$ are merged. The way we produce the cancelling summand for $d_i B_{p,q,d,r\underline{c}}(\underline{\alpha})$ depends only on the row group $i$ and the row group of $i+1$ in the table of the row groups of $B_{p,q,d,r\underline{c}}(\underline{\alpha})$ given in Definition 3.3. It thus suffices to check the following cases, where by case $(pq)$ we mean the case where $i$ lies in group $p$ and $i+1$ lies in group $q$:

- (12): This is the case $i = 1$. It has two subcases. If $c_1 = 1$, the the cancelling summand is
$$B_{p-1,q,d-1,r-1,(c_2,\ldots,c_r)}(d_1\underline{\alpha}).$$
If $c_1 > 1$, then it is
$$d_{m+1}\tau B_{p,q,d,r,(c_2,\ldots,c_r,c_1-1)}(\underline{\alpha}) \quad \text{with} \quad \tau = ((m+1)m\ldots(r+1)).$$
- (22): There are two subcases. If $c_i = c_{i+1}$, the cancelling summand is
$$B_{p-1,q,d-1,r-1,(c_1,\ldots,c_i,c_{i+2},\ldots,c_r)}(d_{\sigma_{\underline{c}}(i)}\underline{\alpha}).$$
If $c_i \neq c_{i+1}$, then it is
$$d_i B_{p,q,d,r,(c_1,\ldots,c_{i-1},c_{i+1},c_i,c_{i+2}\ldots,c_r)}(\underline{\alpha}).$$
- (23): This is the case $i = r+1$. The cancelling summand is
$$d_{r+1}\tau B_{p,q,d,r,\underline{c}}(\underline{\alpha}) \quad \text{with} \quad \tau = ((r+1)(r+2)).$$
- (33): The cancelling summand is
$$B_{p-1,q,d-1,r,\underline{c}}(d_i\underline{\alpha}).$$
- (34): This is the case $i = d$ The cancelling summand is
$$d_d\upsilon B_{p,q,d,r,\underline{c}}(\underline{\alpha}) \quad \text{with} \quad \upsilon = (d(d+1)).$$
- (44): The cancelling summand is
$$B_{p,q,d,r,\underline{c}}(d_{i+p-d}\underline{\alpha}).$$
- (45): This is the case $i = m-p+d$. The cancelling summand is
$$B_{p,q,d,r,\underline{c}}(d_m\underline{\alpha}).$$
- (56): This is the case $i = m-p+d+1$. The cancelling summand is
$$d_{m-p+d+1}\upsilon B_{p,q,d+1,r,\underline{c}}(\underline{\alpha}) \quad \text{with} \quad \upsilon = ((m-p+d+1)\ldots(d+1)).$$
- (66): The cancelling summand is
$$B_{p-1,q,d,r,\underline{c}}(d_{i-m+p-1}\underline{\alpha}).$$
- (61): This is the case $i = m+1$. The cancelling summand is
$$d_1\tau\upsilon B_{p-1,q,d,r,\underline{c}}(\underline{\alpha}) \quad \text{with} \quad \tau = (2\ldots(r+2)), \upsilon = ((r+2)\ldots(d+1)).$$

$p, q, d, r$, and $\underline{c}$ non generic:

These are the cases when one or more of the six row ranges listed in the table defininig $B_{p,q,d,r\underline{c}}(\underline{\alpha})$ in Defn. 3.3 are not present.

This adds to the above analysis the following additional cases where we need to describe the cancelling summand:

- (13): This is the case $r = 0$ but $d \geq 2$. There are two subcases. If $q = n$, the the cancelling summand is
$$B_{p-1,n,d-1,0,\emptyset}(d_1\underline{\alpha}),$$
where $\emptyset$ is the value of $\underline{c}$ when $r = 0$. If $q < n$, the cancelling summand is
$$d_{m+1}\tau B_{p,q,d,1,(n-q)}(\underline{\alpha}) \quad \text{with} \quad \tau = ((m+1)m\dots 2)\,.$$
- (14): This is the case $r = 0$, $d = 1$, $p < m$. The cancelling summand is
$$B_{p+1,q,1,0,\emptyset}(d_{m+1}\underline{\alpha}).$$
- (15): This is the case $r = 0$, $d = 1$, $p = m$ The are two cases. If $q = n$, then the cancelling summand is $\underline{\alpha}$ itself on the RHS of (3.10). If $q \neq n$, then the cancelling summand is
$$B_{1,q+1,1,0,\emptyset}(d_{m+1}\underline{\alpha}).$$
- (24): This is the case $d > 1$, $r = d-1$, $p < m$. The cancelling summand is
$$d_d\tau B_{p,q,d,d-1,\underline{c}}(\underline{\alpha}) \quad \text{with} \quad \tau = (d(d+1))\,.$$
- (25): This is the case $d > 1$, $r = d-1$, $p = m$. The cancelling summand is
$$d_d\tau B_{m,q,d,d-1,\underline{c}}(\underline{\alpha}) \quad \text{with} \quad \tau = (d(d+1))\,.$$
- (35): This is the case $d > 1$, $r < d-1$, $p = m$. The cancelling summand is:
$$d_d B_{m,q,d-1,r,\underline{c}}(\underline{\alpha}).$$
- (51): This is the case $d = p$. There are two subcases. If $q = 2$, the cancelling summand is the summand of $g_n f'_n(\underline{\alpha})$ corresponding to the summand of $g_n$ indexed by
$$\underline{c}' = (1, c_1, \dots, c_r, n-1, \dots, n-1, n, \dots, n) \in \{1, \dots, n\}^m\,, \tag{3.12}$$
where there are $p-1-r$ and $m-p$ instances of $n-1$ and $n$, respectively. If $q \neq 2$, the cancelling summand is
$$d_1\tau B_{m,q-1,d,r,(c_1,\dots,c_r,n-q+1,\dots,n-q+1)}(\alpha) \quad \text{with} \quad \tau = (23\dots(d+1)).$$

Finally, let us assume that $\upsilon$ is non-trivial. Then $d\,(\upsilon B_{p,q,d,r,\underline{c}}(\underline{\alpha}))$ has three kinds of summands. First are those where the differential merges two rows belonging one each to the two row subsets which $\upsilon$ shuffles. Then let $\upsilon'$ be the composition of $\upsilon$ with the transposition interchanging these two rows. Then $d\,(\upsilon' B_{p,q,d,r,\underline{c}}(\underline{\alpha}))$ has an identical summand coming from merging the same two rows. All the sign twists in (3.10) stay the same, except for $(-1)^{\upsilon'} = -(-1)^{\upsilon}$. We conclude that the two summands cancel each other out.

The second type of summands in $d\,(\upsilon B_{p,q,d,r,\underline{c}}(\underline{\alpha}))$ are those which do not belong to the first group, but where the two rows being merged were also adjacent in $B_{p,q,d,r,\underline{c}}(\underline{\alpha})$. Thus there is a summand of $d(B_{p,q,d,r,\underline{c}}(\underline{\alpha}))$ where these two rows are merged. The analysis above give the cancelling counterpart $X$ for this summand. Let $\upsilon' \in S_m$ be the permutation obtained from $\upsilon$ where we treat the two

merged rows as a single row. Then $\upsilon' X$ would cancel the desired summand of $d\left(\upsilon B_{p,q,d,r,\underline{c}}(\underline{\alpha})\right)$. It remains to show that $\upsilon' X$ exists as a summand in (3.11). We verify this by analysing every applicable cancelling summand in the analysis above for the case $\tau = \upsilon = \mathrm{Id}$. Firstly, when $X = B_\bullet(d_\bullet \underline{\alpha})$, it is clear that $\upsilon' X$ is also a summand of $\Phi'_n(d\alpha)$. The remaining cases are:

- (12), the subcase $c_1 > 1$. Then
$$\upsilon' X = \upsilon' d_{m+1}\tau B_{p,q,d,r,(c_2,\ldots,c_r,c_1-1)}(\underline{\alpha}) = d_{m+1}\tau\upsilon B_{p,q,d,r,(c_2,\ldots,c_r,c_1-1)}(\underline{\alpha})$$
where $\tau = ((m+1)m\ldots(r+1))$.
- (22), the subcase $c_i \neq c_{i+1}$. Then
$$\upsilon' X = \upsilon' d_i B_{p,q,d,r,(c_1,\ldots,c_{i-1},c_{i+1},c_i,c_{i+2}\ldots,c_r)}(\underline{\alpha}) = d_i\upsilon B_{p,q,d,r,(c_1,\ldots,c_{i-1},c_{i+1},c_i,c_{i+2}\ldots,c_r)}(\underline{\alpha}).$$
- (23). Then
$$\upsilon' X = \upsilon' d_{r+1}\tau B_{p,q,d,r,\underline{c}}(\underline{\alpha}) = d_{r+1}\tau\upsilon B_{p,q,d,r,\underline{c}}(\underline{\alpha}),$$
where $\tau = ((r+1)(r+2))$.
- (56). Then
$$\upsilon' X = \upsilon' d_{m-p+d+1}\upsilon'' B_{p,q,d+1,r,\underline{c}}(\underline{\alpha}) = d_{m-p+d+1}\upsilon\upsilon'' B_{p,q,d+1,r,\underline{c}}(\underline{\alpha})$$
where $\upsilon'' = \left((m-p+d+1)\ldots(d+1)\right)$.
- (61). Then
$$\upsilon' X = \upsilon' d_1\tau\upsilon'' B_{p-1,q,d,r,\underline{c}}(\underline{\alpha}) = d_1\tau(t_{m+1}\upsilon t_{m+1}^{-1}\upsilon'')B_{p-1,q,d,r,\underline{c}}(\underline{\alpha})$$
where $\tau = \left(2\ldots(r+2)\right), \upsilon'' = \left((r+2)\ldots(d+1)\right), t_{m+1} = (1\ldots m+1)$.
- (13), the subcase $q < n$. Then
$$\upsilon' X = \upsilon' d_{m+1}\tau B_{p,q,d,1,(n-q)}(\underline{\alpha}) = d_{m+1}\upsilon\tau B_{p,q,d,1,(n-q)}(\underline{\alpha}),$$
where $\tau = ((m+1)m\ldots 2)$.
- (51). In the subcase $q = 2$, recall that $X$ is the summand of $g_n f'_n(\underline{\alpha})$ corresponding to the summand of $g_n$ indexed by $\underline{c}'$ as in (3.12). Then $\upsilon' X$ is the summand of $g_n f'_n(\underline{\alpha})$ corresponding to $\upsilon'\underline{c}'$.
  In the subcase $q \neq 2$,
$$\begin{aligned}\upsilon' X &= \upsilon' d_1\tau B_{m,q-1,d,r,(c_1,\ldots,c_r,n-q+1,\ldots,n-q+1)}(\underline{\alpha}) = \\ &= d_1\tau' B_{m,q-1,d,r,(c_1,\ldots,c_r,n-q+1,\ldots,n-q+1)}(\underline{\alpha}),\end{aligned}$$
where $\tau = (23\ldots(d+1))$ and $\tau' = t_{m+1}\upsilon t_{m+1}^{-1}\tau$.

The third and the last type of summands are those which do not belong to the first group and where the two rows being merged were not adjacent in $B_{p,q,d,r,\underline{c}}(\underline{\alpha})$. This necessarily means that one of the rows being merged is involved in the shuffle $\upsilon$, and the other isn't. There are three such cases, where by the case $(ij)$ we mean the case where the rows being merged, before they were permuted by $\upsilon$, belonged to the row groups $i$ and $j$ of the table in Definition 3.3:

- (24): This occurs when the shuffle $\upsilon$ permutes $d+1$-st row (the top row in group 4) just below the $r+1$-st row (the bottom row of the group 2). The cancelling summand to $d_{r+1}\left(\upsilon B_{p,q,d,r,\underline{c}}(\underline{\alpha})\right)$ is then
$$d_{r+1}\left(\tau\upsilon B_{p,q,d,r,\underline{c}}(\underline{\alpha})\right) \qquad \text{with } \tau = ((r+1)(r+2)).$$

- (35): This occurs when $\upsilon$ shuffles the bottom row of the 3rd row group all the way down to the row $m-p+d+1$, just above the 5th row group. The cancelling summand is

$$\upsilon d_{d+m-p} B_{p,q,d-1,r,\underline{c}}(\underline{\alpha}).$$

  where $\upsilon' \in S_m$ is obtained from $\upsilon \in S_{m+1}$ by skipping $d$ in $\{1,\dots,m+1\}$ on the source side, and $\upsilon(d)$ on the target side.
- (14): This occurs when the 2nd row group is empty, 3rd row group is present, but $\upsilon$ shuffles the top row of the 4th group all the way up to the second row. The cancelling summand is

$$\upsilon' B_{p+1,q,d,r,\emptyset}(d_{m+1}\underline{\alpha}),$$

  where $\upsilon' \in S_m$ is obtained from $\upsilon \in S_{m+1}$ by skipping $d+1$ in $\{1,\dots,m+1\}$ on the source side, and $\upsilon(d+1)$ on the target side.

Similar analysis reduces the case of arbitrary $\tau$ to the case $\tau = \mathrm{Id}$. □

3.5. **Homotopy $\Psi$.** We first need to introduce some integer-valued functions:

**Definition 3.5.** For any integers $k, l \geq 1$ define:

$$N_{k,l} := \sum_{c \in \{1,\dots,l\}^k} (-1)^{\sigma_c}. \tag{3.13}$$

where the permutation $\sigma_c \in S_k$ is as in Definition 3.2.

In other words, $N_{k,l}$ is the number of the ways to choose a single entry in each row of a $k \times l$ matrix when counted with the parity determined by the sign of the induced permutation of the rows. The permutation is obtained by travelling down the columns from left to right, taking the order in which we encounter the chosen entries, and ordering their rows accordingly.

**Lemma 3.6.**

$$N_{k,l} = \sum_{i=1}^{l} \sum_{j=1}^{k} N_{k-j,i-1} Y_{j,k-j}.$$

*Proof.* The summands of the RHS each count the (signed) number of the choices of a single entry in each row of a $k \times l$ matrix where there are exactly $j$ chosen entries in the $i$-th column and no entries in columns $i+1,\dots,l$. □

**Lemma 3.7.** *For odd $k$ we have $N_{k+1,l} = N_{k,l}$.*

*Proof.* Let us prove this by double induction on $k, l$. For all $l$, we have $N_{1,l} = l$. To compute $N_{2,l}$ let us sum over the position $m$ of the entry in the first row:

$$N_{2,l} = \sum_{m=1}^{l} (l-m+1) - (m-1) = \sum_{m=1}^{l} l + 2 - 2m = l(l+2) - 2\frac{l(l+1)}{2} = l.$$

Suppose we have the statement for all $N_{i,j}$ for $i \le k$, $j \le l$, and $i+j < k+l$. Let $k = 2s+1$. Then

$$\begin{aligned}
N_{2s+2,l} &= \sum_{i=1}^{l}\sum_{j=1}^{2s+2} N_{2s+2-j,i-1}Y_{j,2s+2-j} \\
&= \sum_{i=1}^{l}\sum_{t=1}^{s} N_{2s+2-2t,i-1}Y_{2t,2s+2-2t} + 1 \qquad \text{(since } Y_{a,b}=0 \text{ for odd } a,b\text{)} \\
&= \sum_{i=1}^{l}\sum_{t=1}^{s} N_{2s+2-2t,i-1}(Y_{2t-1,2s+2-2t} + Y_{2t,2s+2-2t-1}) + 1 \\
&= \sum_{i=1}^{l}\sum_{t=1}^{s} N_{2s+1-(2t-1),i-1}Y_{2t-1,2s+2-2t} + \sum_{i=1}^{l}\sum_{t=1}^{s} N_{2s+1-2t,i-1}Y_{2t,2s+2-2t-1} + 1 \\
&= N_{2s+1,l}.
\end{aligned}$$

□

**Definition 3.8.** Let $n,m,p,q \in \mathbb{Z}$ with $n,m,p \ge 1$, $q \ge 0$, $p+q \le m$.
We define

$$T^{p,q}_{m,n} := \frac{1}{n}(-1)^{(p-1)(m-p-q)}Y_{p-1,q}\sum_{s=1}^{n} N_{p-1,s-1}N_{q,n-s}.$$

**Lemma 3.9.** *Let $n,m,p,q \in \mathbb{Z}$ with $n,m,p \ge 1$, $q \ge 0$, $p+q \le m$. Then:*

*(1) $T^{p,q}_{m,n} = (-1)^{p-1}T^{p,q}_{m-1,n}$.*
*(2) For even $p$ and odd $q$, we have $T^{p,q}_{m,n} = 0$.*
*(3) For odd $p > 1$ and even $q > 0$, we have*

$$T^{p,q}_{m,n} = (-1)^m T^{p-1,q}_{m,n} + T^{p,q-1}_{m,n} \tag{3.14}$$

*Proof.* For (1), note that the only thing that depends on $m$ in the definition of $T^{p,q}_{m,n}$ is the sign twist $(-1)^{(p-1)(m-p-q)}$. For (2), note that by definition $Y_{p-1,q} = 0$ for even $p$ and odd $q$. Finally, (3) follows from Lemma 3.7 and the formula (2.7). □

**Proposition 3.10.** *Let $m,n \in \mathbb{Z}$ with $m,n \ge 1$. Let*

$$\underline{\alpha} := \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \dots \\ \alpha_m \end{pmatrix} \in \mathrm{HC}_{m-1}(\mathcal{A})$$

*be any Hochschild chain of length $m$.*
*Let*

$$\mathrm{HC}_\bullet(\mathcal{A}^{\otimes n};t)_t \underset{g_n}{\overset{f_n}{\rightleftarrows}} \mathrm{HC}_\bullet(\mathcal{A}). \tag{3.15}$$

*be the maps defined in Definitions 3.2 and 3.1, respectively. Then*

$$f_n g_n(\underline{\alpha}) = \sum_{\substack{1\le p\le m \\ 0\le q\le m-p}} T^{p,q}_{m,n}\,\mathrm{Sh}\left(\begin{pmatrix} \alpha_{m-q+1}\dots\alpha_m\alpha_1\dots\alpha_p \\ \alpha_{p+1} \\ \dots \\ \dots \\ \dots \\ \alpha_{m-q} \end{pmatrix}, \begin{pmatrix} 1 \\ \dots \\ p+q \\ \textit{in total} \\ \dots \\ 1 \end{pmatrix}\right). \tag{3.16}$$

*Proof.* By definition, the summands of $g_n(\alpha)$ are enumerated by $c \in \{1, \dots, n\}^m$ with $c_1 = 1$. We think of $c$ as a choice of one entry in each row of an $m \times n$ matrix. The corresponding summand is this matrix with $\alpha_1, \dots, \alpha_n$ placed in the chosen entries in the cyclic order (3.3). All the other entries are filled with identity maps. The sign of this summand is the sign of the permutation $\sigma_c$ defined by setting $\sigma_c(i)$ to be the index of $\alpha_\bullet$ which was placed in row $i$, cf. §3.3.

Applying $f_n$ to such matrix yields a sum indexed by its columns. For each column, we take the other columns of the matrix, compose them in the cyclic order (3.3) and pre-compose them with the top entry of the column. There is no sign change, thus all $m$ summands have the sign $(-1)^{\sigma_c}$, but gain a coefficient $\frac{1}{n}$.

It follows that the summands of $f_n g_n(\alpha)$ on the LHS are precisely those in the sum on the RHS. It remains to show that the coefficients match.

Fix any $1 \le p \le m$ and $0 \le q \le m - p$. Let $X$ be any summand in the corresponding shuffle product on the RHS. Take any choice of $c \in \{1, \dots, n\}^m$ and of a column $1 \le s \le m$ which contribute to $X$ on the LHS. Permuting the elements $c_2, \dots, c_m$ by any permutation $\tau \in S_{m-1}$ which preserves the relative order of $c_i = s$ and the relative order of $c_i \neq s$ produces a new choice $c'$ of entries in the rows with $\sigma'_c = \tau\sigma_c$. On the LHS the summand $c'$ and $s$ contribute to is $\tau(X)$ and the sign of the contribution differs by $(-1)^\tau$ from that of $c$ and $s$. Hence on the LHS the coefficitients of $X$ and $\sigma(X)$ differ by $(-1)^\tau$. On the other hand, the same holds on the RHS by the definition of the shuffle product.

It is therefore enough to show that the coefficient of any one summand in the shuffle product on the RHS matches its coefficient on the LHS.

Let $X$ be the summand in the shuffle product which has $p - 1$ identity maps above $\alpha_{p+1}, \dots, \alpha_{m-q}$ and has the remaining $q$ identity maps below them. Take any $c \in \{1, \dots, n\}^m$ and $1 \le s \le m$ which contribute to $X$ on the LHS. This means that $c_{p+1} = \dots = c_{m-q} = s$ and there are $p$ elements $c_i$ with $c_i < s$ and $q$ with $c_i > s$. Now let $S_{p-1+q} < S_{m-1}$ be the subgroup which permutes first $p - 1$ and the last $q$ entries and let $\tau \in S_{p-1,q} \subset S_{p-1+q}$ be any shuffle. Then, similar to above, applying $\tau$ to $c_2, \dots, c_m$ produces $c'$ such that $\sigma' = \tau\sigma_c$. Since the swapped rows of the matrix all contain Id in the $s$-th column, on the LHS $c'$ and $s$ contribute to the same summand $X$ but with the sign differing by $(-1)^\tau$.

The action of $S_{p-1,q}$ on $c$ has no stabiliser and its orbit contains exactly one element $c'$ where $c'_1, \dots, c'_p < s$ and $c'_{m-q+1}, \dots, c'_q > s$. It follows that to count the contribution to $X$ on the LHS from all $c \in \{1, \dots, n\}^m$ and fixed $s$, we can count contributions only from $c$ with $c_1, \dots, c_p < s$ and $c_{m-q+1}, \dots, c_q > s$ and multiply the result by $Y_{p-1,q} = \sum_{\tau \in S_{p-1,q}} (-1)^\tau$.

It remains to count the contributions to $X$ from $c$ and $s$ with

- $c_1 = 1$,
- $c_{p+1} = \dots = c_{m-q} = s$,
- $1 \le c_2, \dots, c_p \le s - 1$,
- $s + 1 \le c_{m-q+1}, \dots, c_q \le n$.

In terms of choosing the entries in the rows of a $m \times n$ matrix, this means that our chosen entries must lie in four submatrices of size $1 \times 1$, $(p-1) \times (s-1)$, $(m-p-q) \times 1$ and $q \times (n-s)$. We are free to choose any placement of entries within each submatrix. The cyclic order (3.3) traverses the four submatrices sequentially: first all the entries of the first submatrix, then all the entries of the second, etc. This implies that $\sigma_c$ is the product of the four permutations corresponding to each

submatrix. Thus all possible $c$ and $s$, counted with the sign $(-1)^{\sigma_c}$ contribute

$$\frac{1}{n}\sum_{s=1}^{n} N_{1,1}N_{p-1,s-1}N_{s,1}N_{q,n-s} = \frac{1}{n}\sum_{s=1}^{n} N_{p-1,s-1}N_{q,n-s}.$$

Multiplying by $Y_{p-1,q}$ we obtain the total coefficient of $X$ on the LHS. By inspection, this matches the coefficient $T^{p,q}_{m,n}(-1)^{(p-1)(m-p-q)}$ of $X$ on the RHS. Here $(-1)^{(p-1)(m-p-q)}$ is the sign with which $X$ appears in the shuffle product. □

This leads us to define the desired homotopy $\Psi$ as follows:

**Definition 3.11.** For each $n \geq 1$ define a degree $-1$ map

$$\Psi_n \colon \mathrm{HC}_\bullet(\mathcal{A}) \to \mathrm{HC}_\bullet(\mathcal{A})$$

to be the map which sends any chain

$$\underline{\alpha} := \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \dots \\ \alpha_m \end{pmatrix} \in \mathrm{HC}_{m-1}(\mathcal{A})$$

to

$$(-1)^m \sum_{\substack{1\leq p\leq m \\ 0\leq q\leq m-p \\ p+q \text{ even}}} T^{p,q}_{m,n}\,\mathrm{Sh}\left(\begin{pmatrix} \alpha_{m-q+1}\dots\alpha_m\alpha_1\dots\alpha_p \\ \alpha_{p+1} \\ \dots \\ \dots \\ \dots \\ \alpha_{m-q} \end{pmatrix}, \begin{pmatrix} 1 \\ \dots \\ p+q+1 \\ \text{in total} \\ \dots \\ 1 \end{pmatrix}\right). \tag{3.17}$$

**Theorem 3.2.** *We have*

$$d\Psi_n = f_n g_n - \mathrm{Id}\,.$$

*Proof.* For any $m \geq 1$ take any Hochschild chain of length $m$.

$$\underline{\alpha} := \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \dots \\ \alpha_m \end{pmatrix} \in \mathrm{HC}_{m-1}(\mathcal{A})$$

The shuffle product satisfies Leibnitz rule. Moreover, for any $1 \leq p \leq m$ and $0 \leq q \leq m-p$ with $p+q$ even, we have

$$d\begin{pmatrix} \alpha_{m-q+1}\dots\alpha_m\alpha_1\dots\alpha_p \\ \alpha_{p+1} \\ \dots \\ \dots \\ \dots \\ \alpha_{m-q} \end{pmatrix} = \begin{pmatrix} \alpha_{m-q+1}\dots\alpha_m\alpha_1\dots\alpha_{p+1} \\ \alpha_{p+2} \\ \dots \\ \dots \\ \dots \\ \alpha_{m-q} \end{pmatrix} +$$

$$+ \sum_{j=1}^{m-p-q-1} (-1)^j \begin{pmatrix} \alpha_{m-q+1}\dots\alpha_m\alpha_1\dots\alpha_p \\ \alpha_{p+1} \\ \dots \\ \alpha_{p+j}\alpha_{p+j+1} \\ \dots \\ \alpha_{m-q} \end{pmatrix} + (-1)^m \begin{pmatrix} \alpha_{m-q}\dots\alpha_m\alpha_1\dots\alpha_p \\ \alpha_{p+1} \\ \dots \\ \dots \\ \dots \\ \alpha_{m-q-1} \end{pmatrix}$$

and

$$d\begin{pmatrix} 1 \\ \dots \\ p+q+1 \\ \text{in total} \\ \dots \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ \dots \\ p+q \\ \text{in total} \\ \dots \\ 1 \end{pmatrix}$$

It follows that that $d(\Psi_n(\underline{\alpha}))$ is the sum of the following summands:

- For all $1 \leq p \leq m$ and $0 \leq q \leq m-p$

$$\mathrm{Sh}\left(\begin{pmatrix} \alpha_{m-q+1} \dots \alpha_m \alpha_1 \dots \alpha_p \\ \alpha_{p+1} \\ \dots \\ \dots \\ \dots \\ \alpha_{m-q} \end{pmatrix}, \begin{pmatrix} 1 \\ \dots \\ p+q \\ \text{in total} \\ \dots \\ 1 \end{pmatrix}\right)$$

  with the coefficient

$$\begin{cases} T^{p,q}_{m,n} & p+q \text{ even} \\ (-1)^m T^{p-1,q}_{m,n} + T^{p,q-1}_{m,n} & p+q \text{ odd} . \end{cases}$$

- For all $1 \leq p \leq m$, $0 \leq q \leq m-p$ and $1 \leq j \leq m-p-q-1$

$$\mathrm{Sh}\left(\begin{pmatrix} \alpha_{m-q+1} \dots \alpha_m \alpha_1 \dots \alpha_p \\ \alpha_{p+1} \\ \dots \\ \alpha_{p+j}\alpha_{p+j+1} \\ \dots \\ \alpha_{m-q} \end{pmatrix}, \begin{pmatrix} 1 \\ \dots \\ p+q+1 \\ \text{in total} \\ \dots \\ 1 \end{pmatrix}\right)$$

  with the coefficient

$$\begin{cases} (-1)^{m+j}\, T^{p,q}_{m,n} & p+q \text{ even} \\ 0 & p+q \text{ odd} . \end{cases}$$

On the other hand,

$$d\underline{\alpha} = d\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \dots \\ \dots \\ \dots \\ \alpha_m \end{pmatrix} = \sum_{j=1}^{m-1} (-1)^{j-1} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \dots \\ \alpha_j\alpha_{j+1} \\ \dots \\ \alpha_m \end{pmatrix} + (-1)^{m-1} \begin{pmatrix} \alpha_m\alpha_1 \\ \alpha_2 \\ \dots \\ \dots \\ \dots \\ \alpha_m \end{pmatrix}, \tag{3.18}$$

and so $\Psi_n(d\underline{\alpha})$ is the sum of the following summands:

- For all $1 \leq p \leq m$ and $0 \leq q \leq m-p$

$$\mathrm{Sh}\left(\begin{pmatrix} \alpha_{m-q+1} \dots \alpha_m \alpha_1 \dots \alpha_p \\ \alpha_{p+1} \\ \dots \\ \dots \\ \dots \\ \alpha_{m-q} \end{pmatrix}, \begin{pmatrix} 1 \\ \dots \\ p+q \\ \text{in total} \\ \dots \\ 1 \end{pmatrix}\right)$$

with the coefficient

$$\begin{cases} (-1)^{m-1}\, T^{p-1,q}_{m-1,n} + T^{p,q-1}_{m-1,n} & p \text{ even}, q \text{ odd}\ , \\ 0 & \text{otherwise}\ . \end{cases}$$

- For all $1 \le p \le m$, $0 \le q \le m-p$ and $1 \le j \le m-p-q-1$

$$\mathrm{Sh}\left(\begin{pmatrix} \alpha_{m-q+1} \dots \alpha_m \alpha_1 \dots \alpha_p \\ \alpha_{p+1} \\ \dots \\ \alpha_{p+j}\alpha_{p+j+1} \\ \dots \\ \alpha_{m-q} \end{pmatrix}, \begin{pmatrix} 1 \\ \dots \\ p+q+1 \\ \text{in total} \\ \dots \\ 1 \end{pmatrix}\right)$$

with the coefficient

$$\begin{cases} (-1)^{m+p+j}\, T^{p,q}_{m-1,n} & p+q \text{ even} \\ 0 & p+q \text{ odd}\ . \end{cases}$$

The desired assertion now follows by adding the coefficients of the summands in $d(\Psi_n(\underline{\alpha}))$ and $\Psi_n(d\underline{\alpha})$ to obtain the coefficients in $(d\Psi_n)(\underline{\alpha})$ and then applying Lemma 3.9. □

## 4. Noncommutative orbifold decomposition

Let $\mathcal{A}$ be a small DG category. In this section, we use the mutually inverse homotopy equivalences

$$\mathrm{HC}_\bullet(\mathcal{A}^{\otimes n}; t_n)_{t_n} \underset{g}{\overset{f}{\rightleftarrows}} \mathrm{HC}_\bullet(\mathcal{A}).$$

constructed explicitly in §3 to define (also explicitly) the noncommutative orbifold decomposition:

$$\mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A}) \simeq \bigoplus_{\underline{\lambda}\vdash n} \mathrm{Sym}^{r(\underline{\lambda})}\, \mathrm{HH}_\bullet(\mathcal{A}), \tag{4.1}$$

where our notation $\mathrm{Sym}^{r(\underline{\lambda})}$ for symmetric powers of a graded vector space is explained in §2.4.

We construct the decomposition (4.1) in two steps.

4.1. **From the Hochschild homology of a quotient stack to the group-twisted Hochschild homologies.** The first step is very general and works for any quotient stack (under a strong group action), not just the symmetric one.

**Definition 4.1.** Let $\mathcal{A}$ be a small DG category with a strong action of a finite group $G$, cf. §2.2. Define maps

$$\mathrm{HC}_\bullet(\mathcal{A} \rtimes G) \underset{\xi}{\overset{\nu}{\rightleftarrows}} \left(\bigoplus\nolimits_{g\in G} \mathrm{HC}_\bullet(\mathcal{A}; g)\right)_G \tag{4.2}$$

as follows. The map $\nu$ sends

$$(\alpha_0, \sigma_0) \otimes \dots \otimes (\alpha_m, \sigma_m) \in \mathrm{HC}_\bullet(\mathcal{A} \rtimes G)$$

$$\downarrow$$

$$(\sigma_0 \dots \sigma_m)^{-1}(\alpha_0) \otimes (\sigma_1 \dots \sigma_m)^{-1}(\alpha_1) \otimes \dots \otimes (\sigma_m)^{-1}(\alpha_m) \in \mathrm{HC}_m(\mathcal{A}^n; (\sigma_0 \dots \sigma_m)^{-1}).$$

It is a generalisation of [Bar03, Proposition 3.5]. The map $\xi$ is the composition

$$\Big(\bigoplus_{g\in G}\mathrm{HC}_\bullet(\mathcal{A};g)\Big)_G \to \bigoplus_{g\in G}\mathrm{HC}_\bullet(\mathcal{A};g) \xrightarrow{\sum_{g\in G}\xi_g} \mathrm{HC}_\bullet(\mathcal{A}\rtimes G) \tag{4.3}$$

where the first map is

$$[\underline{\alpha}] \mapsto \frac{1}{|G|}\sum_{h\in G} h.\underline{\alpha} \tag{4.4}$$

and each

$$\xi_g\colon\ \mathrm{HC}_\bullet(\mathcal{A};g) \to \mathrm{HC}_\bullet(\mathcal{A}\rtimes G)$$

is the map

$$\alpha_0\otimes\cdots\otimes\alpha_m \mapsto \big((\mathrm{Id},g^{-1})\circ(\alpha_0,\mathrm{Id})\big)\otimes(\alpha_1,\mathrm{Id})\otimes\cdots\otimes(\alpha_{m-1},\mathrm{Id})\otimes(\alpha_m,\mathrm{Id}). \tag{4.5}$$

**Lemma 4.2.** *The maps $\nu$ and $\xi_g$ in Defn. 4.1 are well defined: they are compatible with the Hochschild differential in the twisted complexes $\mathrm{HC}_\bullet(\mathcal{A}\rtimes G)$, $\Big(\bigoplus_{g\in G}\mathrm{HC}_\bullet(\mathcal{A};g)\Big)_G$, and $\mathrm{HC}_\bullet(\mathcal{A};g)$.*

*Proof.* It is immediate that the maps $\nu$ and $\xi_g$ commute with the regular summands of the Hochschild differential. It remains to verify that this also holds for the cyclic permutation summand $d_{cyc}$.

For map $\nu$, we have

$$\begin{aligned}
&\nu\big(d_{cyc}\big((\alpha_0,\sigma_0)\otimes\cdots\otimes(\alpha_{m-1},\sigma_{m-1})\otimes(\alpha_m,\sigma_m)\big)\big) = \\
&= \nu\big((-1)^m(-1)^{\alpha_m|\alpha_0,\cdots\alpha_{m-1}}(\alpha_m,\sigma_m)(\alpha_0,\sigma_0)\otimes\cdots\otimes(\alpha_{m-1},\sigma_{m-1})\big) = \\
&= (-1)^m(-1)^{\alpha_m|\alpha_0,\cdots\alpha_{m-1}}(\sigma_m\sigma_0\ldots\sigma_{m-1})^{-1}(\alpha_m)(\sigma_0\ldots\sigma_{m-1})^{-1}(\alpha_0)\otimes\cdots\otimes\sigma_{m-1}^{-1}(\alpha_{m-1}),
\end{aligned}$$

while

$$\begin{aligned}
&d_{cyc}\nu\big((\alpha_0,\sigma_0)\otimes\cdots\otimes(\alpha_{m-1},\sigma_{m-1})\otimes(\alpha_m,\sigma_m)\big) = \\
&= d_{cyc}\big((\sigma_0\ldots\sigma_m)^{-1}(\alpha_0)\otimes\cdots\otimes(\sigma_{m-1}\sigma_m)^{-1}(\alpha_{m-1})\otimes(\sigma_m)^{-1}(\alpha_m)\big) = \\
&= (-1)^m(-1)^{\alpha_m|\alpha_0,\cdots\alpha_{m-1}}(\sigma_m\sigma_0\ldots\sigma_m)^{-1}(\alpha_m)(\sigma_0\ldots\sigma_m)^{-1}(\alpha_0)\otimes\cdots\otimes(\sigma_{m-1}\sigma_m)^{-1}(\alpha_{m-1}) = \\
&= \sigma_m^{-1}\Big((-1)^m(-1)^{\alpha_m|\alpha_0,\cdots\alpha_{m-1}}(\sigma_m\sigma_0\ldots\sigma_{m-1})^{-1}(\alpha_m)(\sigma_0\ldots\sigma_{m-1})^{-1}(\alpha_0)\otimes\cdots\otimes(\sigma_{m-1})^{-1}(\alpha_{m-1})\Big).
\end{aligned}$$

Thus $d_{cyc}\nu$ and $\nu d_{cyc}$ become equal after taking the $G$-coinvariants.

For each $g\in G$, for map $\xi_g$ we have

$$\begin{aligned}
&\xi_g\left(d_{cyc}\left(\alpha_0\otimes\cdots\otimes\alpha_m\right)\right) = \\
=&\xi_g\left((-1)^m(-1)^{\alpha_m|\alpha_0,\cdots\alpha_{m-1}}g(\alpha_m)\alpha_0\otimes\cdots\otimes\alpha_{m-1}\right) = \\
=&(-1)^m(-1)^{\alpha_m|\alpha_0,\cdots\alpha_{m-1}}\left(\left(\mathrm{Id},g^{-1}\right)\circ\left(g(\alpha_m)\alpha_0,\mathrm{Id}\right)\right)\otimes\cdots\otimes\left(\alpha_{m-1},\mathrm{Id}\right),
\end{aligned}$$

while

$$\begin{aligned}
&d_{cyc}\left(\xi_g\left(\alpha_0\otimes\cdots\otimes\alpha_m\right)\right) = \\
=&d_{cyc}\left(\left(\left(\mathrm{Id},g^{-1}\right)\circ\left(\alpha_0,\mathrm{Id}\right)\right)\otimes\cdots\otimes\left(\alpha_m,\mathrm{Id}\right)\right) = \\
=&\left(\left(\alpha_m,\mathrm{Id}\right)\circ\left(\mathrm{Id},g^{-1}\right)\circ\left(\alpha_0,\mathrm{Id}\right)\right)\otimes\cdots\otimes\left(\alpha_{m-1},\mathrm{Id}\right) = \\
=&(-1)^m(-1)^{\alpha_m|\alpha_0,\cdots\alpha_{m-1}}\left(\left(\mathrm{Id},g^{-1}\right)\circ\left(g(\alpha_m)\alpha_0,\mathrm{Id}\right)\right)\otimes\cdots\otimes\left(\alpha_{m-1},\mathrm{Id}\right).
\end{aligned}$$

Thus $d_{cyc}\xi_g$ and $\xi_g d_{cyc}$ are equal. □

**Proposition 4.3.** *The maps $\nu$ and $\xi$ in Defn. 4.1 are mutually inverse homotopy equivalences and hence induce mutually inverse isomorphisms*

$$\mathrm{HH}_\bullet(\mathcal{A}\rtimes G) \underset{\xi}{\overset{\nu}{\rightleftarrows}} \left(\bigoplus_{g\in G}\mathrm{HH}_\bullet(\mathcal{A};g)\right)_G. \tag{4.6}$$

*Proof.* The composition $\nu\xi$ is the identity map of $\left(\bigoplus_{g\in G}\mathrm{HC}_\bullet(\mathcal{A};g)\right)_G$. The composition $\xi\nu$ is the endomorphism of $\mathrm{HC}_\bullet(\mathcal{A}\rtimes G)$ which sends any chain

$$(\alpha_0,\sigma_0)\otimes(\alpha_1,\sigma_1)\otimes\cdots\otimes(\alpha_m,\sigma_m)\in\mathrm{HC}_m(\mathcal{A}\rtimes G)$$

to the chain

$$\frac{1}{|G|}\sum_{g\in G}(g\alpha_0, g\sigma_0\ldots\sigma_m g^{-1})\otimes(g(\sigma_1\ldots\sigma_m)^{-1}(\alpha_1),\mathrm{Id})\cdots\otimes(g\sigma_m^{-1}(\alpha_m),\mathrm{Id})\in\mathrm{HC}_m(\mathcal{A}\rtimes G).$$

Denote $\beta_i^j=(\sigma_i\ldots\sigma_j)^{-1}\alpha_i$ for $0\le i\le j\le m$. The homotopy between $\xi\nu$ and (4.4) on $\mathrm{HC}_\bullet(\mathcal{A}\rtimes G)$ is given by the degree $-1$ endomorphism which sends any chain

$$(\alpha_0,\sigma_0)\otimes(\alpha_1,\sigma_1)\otimes\cdots\otimes(\alpha_m,\sigma_m)\in\mathrm{HC}_m(\mathcal{A}\rtimes G)$$

to the chain

$$\begin{aligned}\frac{1}{|G|}\sum_{g\in G}g.\Bigg(\sum_{p=0}^{m-1}\sum_{r,(s_1,\ldots,s_r)\vdash m-p}\sum_{J\subset\{p+1,\ldots,m+1\}}(-1)^{(4.7)}(\beta^m_{m-p-r+1}\ldots\beta^m_m\alpha_0,\sigma_0)\otimes\\ \otimes(\alpha_1,\sigma_1)\otimes\ldots\otimes(\alpha_p,\sigma_p)\otimes X_{p+1}(p,\underline{s},J)\otimes\ldots\otimes X_{m+1}(p,\underline{s},J)\Bigg)+\\ +\frac{1}{|G|}\sum_{g\in G}\sum_{i=0}^{m}(-1)^{i-1}(g(\beta_1^m\ldots\beta_m^m),g\sigma_0)\otimes(\mathrm{Id},\sigma_1)\otimes\ldots\otimes(\mathrm{Id},\sigma_i)\otimes(\mathrm{Id},g^{-1})\\ \otimes(\mathrm{Id},g\sigma_{i+1}g^{-1})\otimes\ldots\otimes(\mathrm{Id},g\sigma_m g^{-1}),\end{aligned}$$

where $J$ is a subset of $\{p+1,\ldots,m+1\}$ such that for all $i\in\{p+1,\ldots,m+1\}$ we have $\sum_{j\in J,\ j<i}s_j\ge|\{j\notin J, p<j\le i\}|$; in particular, $p+1\in J$ always. Furthermore, let $k(i)=|\{j\in J,\ j<i\}$, then let

$$X_i(p,s_1,\ldots,s_r,J)=(\mathrm{Id},\sigma_{p+s_1+\ldots s_{k(i)}+1}\ldots\sigma_{p+s_1+\ldots s_{k(i)+1}})$$

if $i\in J$, and

$$X_i(p,s_1,\ldots,s_r,J)=(\beta_{i-k(i)}^{p+s_1+\ldots+s_{k(i)}-i+k(i)},\mathrm{Id})$$

if $i\notin J$. In other words, after $(\alpha_p,\sigma_p)$ we disjoin every $(\alpha_i,\sigma_i)$ into $(\alpha_i,\mathrm{Id})$ and $(\mathrm{Id},\sigma_i)$, commute some $(\mathrm{Id},\sigma_i)$ towards the left according to the relations in $\mathcal{A}\rtimes S_n$, multiply some of them together, and rotate some $(\beta_q^m,\mathrm{Id})$ from the right end into the first term in the chain, all of it so that $m+2$ total terms remain. We think of $(s_1,\ldots,s_r)$ as the ordered partition of the set $\{\sigma_{p+1},\ldots,\sigma_m\}$ and the set $J$ as the set of indices where the products of $(\mathrm{Id},\sigma_i)$ end up. Then $\sigma_0,\ldots,\sigma_{p+s_1+\ldots+s_{k(i)}}$ are the ones that are to the left of the term number $i$.

The sign in the second summation is given by

$$\left(\sum_{i=0}^{m-p-r}\deg\alpha_i\right)\cdot\left(\sum_{i=m-p-r+1}^{m}\deg\alpha_i\right)+(p+m-1)r+\sum_{j\in J}j-p-k(j). \tag{4.7}$$

□

4.2. **From** $\mathrm{HC}_\bullet(\mathcal{A}^{\otimes n};\sigma_{\underline{\lambda}})_{C(\sigma_{\underline{\lambda}})}$ **to** $\mathrm{Sym}^{\underline{r}(\underline{\lambda})}\mathrm{HC}_\bullet(\mathcal{A})$**.** For each conjugacy class $\underline{\lambda}\vdash n$ of $S_n$, choose the representative

$$\sigma_{\underline{\lambda}} := \big(1\dots n_1\big)\big((n_1+1)\dots(n_1+n_2)\big)\dots\big((n_1+\dots+n_{r(\underline{\lambda}-1)}+1)\dots n\big)\in S_n$$

where $n_1\le n_2\le\dots\le n_{r(\underline{\lambda})}$ is the unique ordering of the parts of $\underline{\lambda}$ in non-decreasing order. For example if $\underline{\lambda}=\{1112333344\}$, then

$$\sigma_{\underline{\lambda}} = (1)(2)(3)(45)(678)(9\ 10\ 11)(12\ 13\ 14)(15\ 16\ 17)(18\ 19\ 20\ 21)(22\ 23\ 24\ 25).$$

This (or any other) choice defines an isomorphism

$$\Big(\bigoplus_{\sigma\in S_n}\mathrm{HC}_\bullet(\mathcal{A};\sigma)\Big)_{S_n}\xrightarrow{\sim}\bigoplus_{\underline{\lambda}\vdash n}\mathrm{HC}_\bullet(\mathcal{A}^{\otimes n};\sigma_{\underline{\lambda}})_{C(\sigma_{\underline{\lambda}})},\tag{4.8}$$

where $C(-)$ denotes the centraliser of a group element. This isomorphism is given by the projection to the corresponding direct summands.

We then define:

**Definition 4.4.** Let $\mathcal{A}$ be a small DG category. For any $n\ge 0$ and any $\underline{\lambda}\vdash n$, define the maps

$$\mathrm{HC}_\bullet(\mathcal{A}^{\otimes n};\sigma_{\underline{\lambda}})_{C(\sigma_{\underline{\lambda}})}\underset{g_{\underline{\lambda}}}{\overset{f_{\underline{\lambda}}}{\rightleftarrows}}\mathrm{Sym}^{\underline{r}(\underline{\lambda})}\mathrm{HC}_\bullet(\mathcal{A})\tag{4.9}$$

to be given by the composition

$$\begin{array}{ccccc}\mathrm{HC}_\bullet(\mathcal{A}^{\otimes n};\sigma_{\underline{\lambda}})_{C(\sigma_{\underline{\lambda}})} & \underset{\mathrm{EZ}}{\overset{\mathrm{AW}}{\rightleftarrows}} & \Big(\bigotimes_{i=1}^{r(\underline{\lambda})}\mathrm{HC}_\bullet(\mathcal{A}^{\otimes n_i};t_{n_i})_{t_{n_i}}\Big)_{S_{\underline{r}(\underline{\lambda})}} & \underset{\otimes g_{n_i}}{\overset{\otimes f_{n_i}}{\rightleftarrows}} & \Big(\bigotimes_{i=1}^{r(\underline{\lambda})}\mathrm{HC}_\bullet(\mathcal{A})\Big)_{S_{\underline{r}(\underline{\lambda})}}\\ & & & & \Big\|\\ & & & & \mathrm{Sym}^{\underline{r}(\underline{\lambda})}\mathrm{HC}_\bullet(\mathcal{A}),\end{array}$$

where

- EZ and AW are the Eilenberg-Zilber and the Alexander-Whitney maps which lift to the level of Hochschild complexes the Künneth isomorphism and its inverse, cf. §2.7. We first apply these without the coinvariants, using $\mathcal{A}^{\otimes n}=\bigotimes_{i=1}^{r(\underline{\lambda})}\mathcal{A}^{n_i}$ and $\sigma_{\underline{\lambda}}=t_{n_1}\times\dots\times t_{n_{r(\underline{\lambda})}}\in S_{n_1}\times\dots\times S_{n_{r(\underline{\lambda})}}<S_n$. To see that they descend to the coinvariant spaces, we observe that EZ and AW intertwine the action of each $t_{n_i}$ on the RHS with the action on the LHS of $t_{n_i}\in S_{n_i}<S_{n_1}\times\dots\times S_{n_{r(\underline{\lambda})}}<S_n$. Furthermore, they intertwine the action of $S_{\underline{r}(\underline{\lambda})}$ on the LHS and with the action on the RHS of $S_{\underline{r}(\underline{\lambda})}<S_n$, the subgroup which permutes the parts of the same size in $n_1,\dots,n_{\underline{r}(\underline{\lambda})}$. Finally, $C(t_{n_1}\times\dots\times t_{n_{r(\underline{\lambda})}})$ is generated by $\langle t_{n_1}\rangle\times\dots\left\langle t_{n_{r(\underline{\lambda})}}\right\rangle$ and $S_{\underline{r}(\underline{\lambda})}$.
- The maps $f_{n_i}$ and $g_{n_i}$ are the mutually inverse homotopy equivalences constructed in §3, cf. §3.2 and §3.3.
- The second vertical equality is due to our notation for the symmetric power indexed by an ordered partition, cf. §2.4.

4.3. **Noncommutative orbifold decomposition.** We now put together all the maps we constructed so far to prove the main theorem of this paper:

**Theorem 4.1** (Noncommutative orbifold decomposition)**.** *Let $\mathcal{A}$ be a small DG category and $n \geq 0$. The following compositions are mutually inverse homotopy equivalences of twisted complexes over* **Mod** $\Bbbk$*:*

$$\begin{array}{ccc} \mathrm{HC}_\bullet(\mathcal{S}^n A) & \underset{\xi}{\overset{\nu}{\rightleftarrows}} & \left(\bigoplus_{\sigma\in S_n} \mathrm{HC}_\bullet(\mathcal{A}^{\otimes n};\sigma)\right)_{S_n} \\ & & {\scriptstyle (4.8)^{-1}}\uparrow \quad \downarrow{\scriptstyle (4.8)} \\ & & \bigoplus_{\underline{\lambda}\vdash n} \mathrm{HC}_\bullet(\mathcal{A}^{\otimes n};\sigma_{\underline{\lambda}})_{C(\sigma_{\underline{\lambda}})} \end{array} \quad \underset{\sum g_{\underline{\lambda}}}{\overset{\sum f_{\underline{\lambda}}}{\rightleftarrows}} \quad \bigoplus_{\underline{\lambda}\vdash n} \mathrm{Sym}^{\underline{r}(\underline{\lambda})} \mathrm{HC}_\bullet(\mathcal{A}), \tag{4.10}$$

*where $\nu$ and $\xi$ are the maps constructed in §4.1 and $f_{\underline{\lambda}}$ and $g_{\underline{\lambda}}$ – in §4.2.*

*In particular, denote the rightward through composition by $\zeta_n$ and the leftward by $\eta_n$. These induce mutually inverse isomorphisms:*

$$\mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A}) \underset{\eta_n}{\overset{\zeta_n}{\underset{\simeq}{\rightleftarrows}}} \bigoplus_{\underline{\lambda}\vdash n} \mathrm{Sym}^{\underline{r}(\underline{\lambda})} \mathrm{HH}_\bullet(\mathcal{A}). \tag{4.11}$$

*Proof.* By Prop. 4.3, the maps $\nu$ and $\xi$ are mutually inverse homotopy equivalences. By construction, the map (4.8) is an isomorphism. Finally, by their construction in Defn. 4.4, the maps $f_{\underline{\lambda}}$ and $g_{\underline{\lambda}}$ are compositions of the Eilenberg-Zilber and the Alexander-Whitney maps EZ and AW with the maps $\otimes f_{n_i}$ and $\otimes g_{n_i}$. The former two are mutually inverse homotopy equivalences lifting the Künneth isomorphism and its inverse, cf. §2.7. The latter two are mutually inverse homotopy equivalences by Theorems 3.1 and 3.2.

Finally, over a field of characteristic 0 taking the coinvariants is an exact functor. Thus passing to the cohomology in (4.10) we obtain (4.11). □

4.4. **Symmetric algebra structure.** In this section, we observe that our noncommutative orbifold decomposition (4.10) defines an explicit isomorphism between the total Hochschild homology $\bigoplus_{n\geq 0} \mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A})$ and the symmetric algebra $S^*\left(\mathrm{HH}_\bullet(\mathcal{A}) \otimes t\Bbbk[t]\right)$. We then use this in §5 to translate some well-known structures which exist on the symmetric algebra to the total Hochschild homology.

We first introduce some notation. Recall that for any $E \in \text{Pre-Tr}^-(\mathbf{Mod}\text{-}\Bbbk)$ and any ordered partition $(n_1, \dots, n_m)$ of $n$ we write

$$\mathrm{Sym}^{n_1,\dots,n_m} E := (E^{\otimes n})_{S_{n_1,\dots,n_m}} \simeq \mathrm{Sym}^{n_1} E \otimes \mathrm{Sym}^{n_2} E \otimes \cdots \otimes \mathrm{Sym}^{n_m} E.$$

View $t\Bbbk[t]$ as a $\Bbbk$-vector space and hence as an object in **Mod** $\Bbbk$. It has a basis $\{t, t^2, t^3, \dots, \}$. Given any $E \in \text{Pre-Tr}^-(\mathbf{Mod}\text{-}\Bbbk)$, we have

$$E \otimes t\Bbbk[t] \simeq \bigoplus_{i\geq 1} E \otimes t^i.$$

Write, for brevity, $Et^i$ for $E \otimes t^i$. We have

$$\begin{aligned} S^*(E \otimes t\Bbbk[t]) := \bigoplus_{m\geq 0} \mathrm{Sym}^m(E \otimes t\Bbbk[t]) \simeq \bigoplus_{m\geq 0} \mathrm{Sym}^m\Big(\bigoplus_{i\geq 1} Et^i\Big) \simeq \\ \simeq \bigoplus_{m\geq 0} \bigoplus_{\substack{m_1,m_2,\dots\in\mathbb{Z}_{\geq 0},\\ \sum m_i = m}} \mathrm{Sym}^{m_1,m_2,m_3,\dots} E \;\; (t^1)^{m_1}(t^2)^{m_2}\dots \end{aligned} \tag{4.12}$$

As we view $t\Bbbk[t]$ only as a vector space, in the symmetric algebra $(t^i)^j \neq t^{ij}$.

We next regroup the summands using the fact the each ordered partition $m_1, m_2, \dots$ of $m$ is $\underline{r}(\underline{\lambda})$ of the unique unordered partition $\underline{\lambda} \vdash n$ where $n = \sum m_i i$ and $\underline{\lambda}$ has

$m_1$ parts of size 1, $m_2$ parts of size 2, etc. For example, in the sum above the ordered partition $(4, 0, 1, 2, 0, \dots)$ of 7 indexes the monomial

$$(t^1)^4 t^3 (t^4)^2 = t^1 t^1 t^1 t^1 t^3 t^4 t^4.$$

We can think of the same monomial as being indexed by the unordered partition $\{1111344\}$ of 15. We thus define for any $n \geq 0$ and any $\underline{\lambda} \vdash n$

$$t^{\underline{\lambda}} := t^{n_1} t^{n_2} \dots t^{n_{r(\underline{\lambda})}},$$

and regroup the above as:

$$\bigoplus_{n \geq 0} \bigoplus_{\underline{\lambda} \vdash n} \left( \mathrm{Sym}^{\underline{r(\lambda)}} E \right) t^{\underline{\lambda}}. \tag{4.13}$$

We introduce the following notation. For any $n \geq 0$ and $\underline{\lambda} \vdash n$, any element of $\mathrm{Sym}^{\underline{r(\lambda)}} E$ can be written as the sum of basic elements

$$\underline{s}_1 \otimes \dots \otimes \underline{s}_n \qquad \underline{s}_i \in \mathrm{Sym}^{\underline{r(\lambda)}} E,$$

and we further write

$$\underline{s}_i = s_{i1} \vee \dots \vee s_{ir_i(\underline{\lambda})} \qquad s_{ij} \in E.$$

Here we distinguish between the skew-commutative product $\vee$ within each $\mathrm{Sym}^{r_i(\underline{\lambda})} E$ and ordinary tensor product $\otimes$ in $\mathrm{Sym}^{r_1(\underline{\lambda})} E \otimes \dots \otimes \mathrm{Sym}^{r_n(\underline{\lambda})} E$.

For example, when $n = 15$ and $\underline{\lambda} = (1111344)$, we have

$$t^{\underline{\lambda}} = t^1 t^1 t^1 t^1 t^3 t^4 t^4,$$

$$\mathrm{Sym}^{\underline{r(\lambda)}} = \mathrm{Sym}^4 E \otimes E \otimes \mathrm{Sym}^2 E,$$

and any element of $\mathrm{Sym}^{\underline{r(\lambda)}} E$ can be written as a sum of the elements of form

$$\underline{s} = (s_{11} \vee s_{12} \vee s_{13} \vee s_{14}) \otimes (s_{31}) \otimes (s_{31} \vee s_{32}).$$

We thus have an isomorphism

$$S^*(E \otimes t\Bbbk[t]) \simeq \bigoplus_{n \geq 0} \bigoplus_{\underline{\lambda} \vdash n} \left( \mathrm{Sym}^{\underline{r(\lambda)}} E \right) t^{\underline{\lambda}}. \tag{4.14}$$

It sends any

$$s_1 t^{n_1} \vee s_2 t^{n_2} \vee \dots \vee s_m t^{n_m} \in S^*(E \otimes t\Bbbk[t])$$

with $n_1 \leq n_2 \leq \dots \leq n_m$ to

$$(s_1 \vee \dots \vee s_{m_1}) \otimes (s_{m_1+1} \vee \dots \vee s_{m_1+m_2}) \otimes \dots \otimes (s_{m_1+\dots+m_{m-1}+1} \vee \dots \vee s_n) t^{\underline{\lambda}}$$

where $n = \sum n_i$, $\underline{\lambda} = \{n_1, \dots, n_m\}$ and $m_i = r_i(\underline{\lambda})$. The inverse isomorphism sends any

$$\underline{s}_1 \otimes \dots \otimes \underline{s}_n t^{\underline{\lambda}} \in \left( \mathrm{Sym}^{\underline{r(\lambda)}} E \right) t^{\underline{\lambda}},$$

to

$$(s_{11} t^1) \vee (s_{12} t^1) \vee \dots \vee (s_{nr_n(\underline{\lambda})} t^n).$$

We thus have the folowing description of the total Hochschild homology of $\mathcal{S}^n \mathcal{A}$:

**Theorem 4.2.** *Let $\mathcal{A}$ be a small DG category. Define the maps*

$$\bigoplus_{n\geq 0} \mathrm{HC}_\bullet(\mathcal{S}^n A) \underset{\eta}{\overset{\zeta}{\rightleftarrows}} S^*(\mathrm{HC}_\bullet(\mathcal{A}) \otimes t\Bbbk[t]) \tag{4.15}$$

*to be the compositions*

$$\bigoplus_{n\geq 0} \mathrm{HC}_\bullet(\mathcal{S}^n A) \underset{\sum_n \eta_n}{\overset{\sum_n \zeta_n}{\rightleftarrows}} \bigoplus_{n\geq 0} \bigoplus_{\underline{\lambda}\vdash n} \mathrm{Sym}^{\underline{r}(\underline{\lambda})} \mathrm{HC}_\bullet(\mathcal{A})\, t^{\underline{\lambda}} \underset{(4.14)}{\overset{(4.14)^{-1}}{\rightleftarrows}} S^*(\mathrm{HC}_\bullet(\mathcal{A}) \otimes t\Bbbk[t])$$

*Then $\zeta$ and $\eta$ are mutually inverse homotopy equivalences. In particular, they induce mutually inverse isomorphims:*

$$\bigoplus_{n\geq 0} \mathrm{HH}_\bullet(\mathcal{S}^n A) \underset{\eta}{\overset{\zeta}{\underset{\simeq}{\rightleftarrows}}} S^*(\mathrm{HH}_\bullet(\mathcal{A}) \otimes t\Bbbk[t]). \tag{4.16}$$

*Proof.* The map (4.14) is an isomorphism, and by Theorem 4.1 the maps $\sum_n \zeta_n$ and $\sum_n \eta_n$ are mutually inverse homotopy equivalences. □

## 5. Hopf algebra and $\lambda$-ring structures

In this section we use explicit isomorphisms $\eta$ and $\zeta$ between $\bigoplus_{n\geq 0} \mathrm{HH}_\bullet(\mathcal{S}^n A)$ and $S^*(\mathrm{HH}_\bullet(\mathcal{A}) \otimes t\Bbbk[t])$ established in §4 to compare – and define – some structures on these spaces.

5.1. **Hopf algebra structure.** For any $\Bbbk$-vector space $E$ its tensor algebra $T^*(E)$ is naturally a Hopf algebra, and the symmetric algebra $S^*(E)$ is its Hopf subalgebra, cf. [Swe69, §4.0]. For $S^*(E)$ the Hopf algebra structure agrees with the universal enveloping algebra of the Lie algebra $E$ with the zero bracket.

We recall the details to introduce the notation:

**Definition 5.1.** Let $\mathcal{A}$ be a small DG category and let $E \in \text{Pre-Tr}^-(\mathcal{A})$. The symmetric algebra $S^*(E)$ is a Hopf algebra with the following operations:

- Multiplication $\mu\colon S^*(E) \otimes S^*(E) \to S^*(E)$ given by
$$(s_1 \vee \cdots \vee s_n) \otimes (t_1 \vee \cdots \vee t_m) \to s_1 \vee \cdots \vee s_n \vee t_1 \vee \cdots \vee t_m,$$
- Comultiplication $\Delta\colon S^*(E) \to S^*(E) \otimes S^*(E)$ given by
$$(s_1 \vee \cdots \vee s_n) \to \sum_{\{i_1,\ldots,i_m\}\subseteq\{1,\ldots,n\}} (-1)^p (s_{i_1} \vee \cdots \vee s_{i_m}) \otimes (s_{j_1} \vee \cdots \vee s_{j_{n-m}}),$$
where $\{j_1,\ldots,j_{n-m}\} = \{1,\ldots,n\} \setminus \{i_1,\ldots,i_m\}$ and the sign twist $(-1)^p$ is the product of $(-1)^{\deg(s_{i_k})\deg(s_{j_l})}$ for each $i_k > j_l$,
- Unit $u\colon \Bbbk \to S^*(E)$ and counit $\epsilon\colon S^*(E) \to \Bbbk$ given by the inclusion of and the projection onto $S^0(E) = \Bbbk$,
- Antipode map $H\colon S^*(E) \to S^*(E)$ given on $S^n(E)$ by $(-1)^n \mathrm{Id}$.

In particular, this defines a Hopf algebra structure on the symmetric algebra $S^*(\mathrm{HH}_\bullet(\mathcal{A}) \otimes t\Bbbk[t])$. We next compute the corresponding Hopf algebra structure on $\bigoplus_{n\geq 0} \mathrm{HH}_\bullet(\mathcal{S}^n A)$:

**Theorem 5.1.** *Let $\mathcal{A}$ be a small DG category. The isomorphisms $\zeta$ and $\eta$ of Theorem 4.2*

$$\bigoplus_{n\geq 0} \mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A}) \underset{\eta}{\overset{\zeta}{\underset{\simeq}{\rightleftarrows}}} S^*(\mathrm{HH}_\bullet(\mathcal{A})\otimes t\Bbbk[t]).$$

*identify the Hopf algebra operations $\mu$, $\Delta$, $u$, $\epsilon$, and $H$ on $S^*(\mathrm{HH}_\bullet(\mathcal{A})\otimes t\Bbbk[t])$ defined in Definition 5.1 with the following operations on $\bigoplus_{n\geq 0}\mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A})$:*

- *The map*

$$\mu'\colon \Big(\bigoplus_{n\geq 0}\mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A})\Big)\otimes\Big(\bigoplus_{n\geq 0}\mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A})\Big)\to\bigoplus_{n\geq 0}\mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A})$$

  *is given by the sum over all $n,m\geq 0$ of the maps*

$$\mathrm{HC}_\bullet(\mathcal{S}^n\mathcal{A})\otimes\mathrm{HC}_\bullet(\mathcal{S}^m\mathcal{A})\xrightarrow{\mathrm{EZ}}\mathrm{HC}_\bullet(\mathcal{S}^n\mathcal{A}\otimes\mathcal{S}^m\mathcal{A})\xrightarrow{\mathrm{Ind}_{S_n\times S_m}^{S_{n+m}}}\mathrm{HC}_\bullet(\mathcal{S}^{n+m}\mathcal{A}). \quad (5.1)$$

- *The map*

$$\Delta'\colon \bigoplus_{n\geq 0}\mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A})\to\Big(\bigoplus_{n\geq 0}\mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A})\Big)\otimes\Big(\bigoplus_{n\geq 0}\mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A})\Big)$$

  *given by the sum over all $n\geq 0$ and all subsets $I\subseteq\{1,\dots,n\}$ of the maps*

$$\begin{array}{ccc}\mathrm{HC}_\bullet(\mathcal{S}^n\mathcal{A}) & \xrightarrow{\mathrm{Res}_{S_I\times S_{\bar I}}^{S_n}} & \mathrm{HC}_\bullet\left(\mathcal{A}^{\otimes n}\rtimes(S_I\times S_{\bar I})\right)\\ & & \big\downarrow\simeq\\ & & \mathrm{HC}_\bullet(\mathcal{S}^{|I|}\mathcal{A}\otimes\mathcal{S}^{n-|I|}\mathcal{A})\xrightarrow{\mathrm{AW}}\mathrm{HC}_\bullet(\mathcal{S}^{|I|}\mathcal{A})\otimes\mathrm{HC}_\bullet(\mathcal{S}^{n-|I|}\mathcal{A}),\end{array}$$

  *where $\bar I=\{1,\dots,n\}\setminus I$, where $S_I, S_{\bar I}<S_n$ are the subgroups that only permute the elements of $I$ and of $\bar I$, and where $|I|$ is the size of $I$.*
- *The maps $u'\colon\Bbbk\to\bigoplus_{n\geq 0}\mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A})$ and $\epsilon'\colon\bigoplus_{n\geq 0}\mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A})\to\Bbbk$ given by the inclusion of and the projection onto $\mathrm{HH}_\bullet(\mathcal{S}^0\mathcal{A})\simeq\Bbbk$,*
- *The map $H'\colon\bigoplus_{n\geq 0}\mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A})\to\bigoplus_{n\geq 0}\mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A})$ given on $\mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A})$ by $(-1)^n\,\mathrm{Id}$.*

*In particular, $\mu'$, $\Delta'$, $u'$, $\epsilon'$ and $H'$ define a structure of a commutative Hopf algebra on $\bigoplus_{n\geq 0}\mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A})$.*

*Proof.* The assertions about the unit map $u$, the counit map $\epsilon$, and the antipode map $H$ being identified with the maps $u'$, $\epsilon'$, and the map $H'$ are trivial.

For the multiplication $\mu$, in view of the isomorphism (4.14) it suffices to prove that for any $m,n\geq 0$, for any $\underline{\nu}\vdash m$ and $\underline{\lambda}\vdash n$, and for any $x\in\mathrm{Sym}^{\underline{\nu}}\mathrm{HH}_\bullet(\mathcal{A})$ and $y\in\mathrm{Sym}^{\underline{\lambda}}\mathrm{HH}_\bullet(\mathcal{A})$ we have

$$\zeta(\mu(xt^{\underline{\nu}}\otimes yt^{\underline{\lambda}}))=\mu'(\zeta(xt^{\underline{\nu}})\otimes\zeta(yt^{\underline{\lambda}}))). \quad (5.2)$$

One of the composants of $\zeta$, the map $\xi$ which was defined in (4.3) to be

$$\Big(\bigoplus_{g\in G}\mathrm{HC}_\bullet(\mathcal{A};g)\Big)_G\xrightarrow{(4.4)}\bigoplus_{g\in G}\mathrm{HC}_\bullet(\mathcal{A};g)\xrightarrow{\sum_{g\in G}\xi_g}\mathrm{HC}_\bullet(\mathcal{A}\rtimes G)$$

can be rewritten as

$$\Big(\bigoplus_{g\in G}\mathrm{HC}_\bullet(\mathcal{A};g)\Big)_G\xrightarrow{\sum_{g\in G}\xi_g}\mathrm{HC}_\bullet(\mathcal{A}\rtimes G)_G\xrightarrow{(4.4)}\mathrm{HC}_\bullet(\mathcal{A}\rtimes G).$$

In view of this, (5.2) is equivalent to the commutation of the following diagrams:

$$\begin{array}{ccc}
\left(\mathrm{HH}_\bullet(\mathcal{A})^{\otimes r(\underline{\nu})}\right)_{S_{r(\underline{\nu})}} \otimes \left(\mathrm{HH}_\bullet(\mathcal{A})^{\otimes r(\underline{\lambda})}\right)_{S_{r(\underline{\lambda})}} & \twoheadrightarrow & \left(\mathrm{HH}_\bullet(\mathcal{A})^{\otimes r(\underline{\nu}+\underline{\lambda})}\right)_{S_{r(\underline{\nu}+\underline{\lambda})}} \\
\downarrow{\scriptstyle (\otimes g_{m_i})\otimes(\otimes g_{n_j})} & & \downarrow{\scriptstyle (\otimes g_{m_i})\otimes(\otimes g_{n_j})} \\
\left(\bigotimes_{\underline{\nu}} \mathrm{HH}_\bullet(\mathcal{A}^{\otimes\bullet};t_\bullet)_{t_\bullet}\right)_{S_{r(\underline{\nu})}} \otimes \left(\bigotimes_{\underline{\lambda}} \mathrm{HH}_\bullet(\mathcal{A}^{\otimes\bullet};t_\bullet)_{t_\bullet}\right)_{S_{r(\underline{\lambda})}} & \twoheadrightarrow & \left(\bigotimes_{\underline{\nu}+\underline{\lambda}} \mathrm{HH}_\bullet(\mathcal{A}^{\otimes\bullet};t_\bullet)_{t_\bullet}\right)_{S_{r(\underline{\nu}+\underline{\lambda})}} \\
\downarrow{\scriptstyle \mathrm{EZ}\otimes\mathrm{EZ}} & & \downarrow{\scriptstyle \mathrm{EZ}} \\
\mathrm{HH}_\bullet(\mathcal{A}^{\otimes m};\sigma_{\underline{\nu}})_{C(\sigma_{\underline{\nu}})} \otimes \mathrm{HH}_\bullet(\mathcal{A}^{\otimes n};\sigma_{\underline{\lambda}})_{C(\sigma_{\underline{\lambda}})} & \xrightarrow{\mathrm{EZ}} & \mathrm{HH}_\bullet(\mathcal{A}^{\otimes m+n};\sigma_{\underline{\nu}}\times\sigma_{\underline{\lambda}})_{C(\sigma_{\underline{\nu}}\times\sigma_{\underline{\lambda}})}
\end{array} \tag{5.3}$$

$$\begin{array}{ccc}
\mathrm{HH}_\bullet(\mathcal{A}^{\otimes m};\sigma_{\underline{\nu}})_{C(\sigma_{\underline{\nu}})} \otimes \mathrm{HH}_\bullet(\mathcal{A}^{\otimes n};\sigma_{\underline{\lambda}})_{C(\sigma_{\underline{\lambda}})} & \xrightarrow{\mathrm{EZ}} & \mathrm{HH}_\bullet(\mathcal{A}^{\otimes m+n};\sigma_{\underline{\nu}}\times\sigma_{\underline{\lambda}})_{C(\sigma_{\underline{\nu}})\times C(\sigma_{\underline{\lambda}})} \\
\downarrow{\scriptstyle \xi_{\sigma_{\underline{\nu}}}\otimes\xi_{\sigma_{\underline{\lambda}}}} & & \downarrow{\scriptstyle \xi_{\sigma_{\underline{\nu}}\times\sigma_{\underline{\lambda}}}} \\
\mathrm{HH}_\bullet(\mathcal{S}^m\mathcal{A})_{S_m} \otimes \mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A})_{S_n} & \xrightarrow{\mathrm{EZ}} & \mathrm{HH}_\bullet(\mathcal{S}^m\mathcal{A}\otimes\mathcal{S}^n\mathcal{A})_{S_m\times S_n} \\
\downarrow{\scriptstyle (4.4)\otimes(4.4)} & & \downarrow{\scriptstyle (4.4)} \\
\mathrm{HH}_\bullet(\mathcal{S}^m\mathcal{A}) \otimes \mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A}) & \xrightarrow{\mathrm{EZ}} & \mathrm{HH}_\bullet(\mathcal{S}^m\mathcal{A}\otimes\mathcal{S}^n\mathcal{A})
\end{array} \tag{5.4}$$

$$\begin{array}{ccc}
\mathrm{HH}_\bullet(\mathcal{A}^{\otimes m+n};\sigma_{\underline{\nu}}\times\sigma_{\underline{\lambda}})_{C(\sigma_{\underline{\nu}})\times C(\sigma_{\underline{\lambda}})} & \twoheadrightarrow & \mathrm{HH}_\bullet(\mathcal{A}^{\otimes m+n};\sigma_{\underline{\nu}}\times\sigma_{\underline{\lambda}})_{C(\sigma_{\underline{\nu}}\times\sigma_{\underline{\lambda}})} \\
\downarrow{\scriptstyle \xi_{\sigma_{\underline{\nu}}\times\sigma_{\underline{\lambda}}}} & & \downarrow{\scriptstyle \xi_{\sigma_{\underline{\nu}}\times\sigma_{\underline{\lambda}}}} \\
\mathrm{HH}_\bullet(\mathcal{S}^m\mathcal{A}\otimes\mathcal{S}^n\mathcal{A})_{S_m\times S_n} & \xrightarrow{\mathrm{Ind}_{S_m\times S_n}^{S_{m+n}}} & \mathrm{HH}_\bullet(\mathcal{S}^{m+n}\mathcal{A})_{S_{m+n}} \\
\downarrow{\scriptstyle (4.4)} & & \downarrow{\scriptstyle (4.4)} \\
\mathrm{HH}_\bullet(\mathcal{S}^m\mathcal{A}\otimes\mathcal{S}^n\mathcal{A}) & \xrightarrow{\mathrm{Ind}_{S_m\times S_n}^{S_{m+n}}} & \mathrm{HH}_\bullet(\mathcal{S}^{m+n}\mathcal{A})
\end{array} \tag{5.5}$$

where:

- $\bigotimes_{\underline{\nu}} \mathrm{HC}_\bullet(\mathcal{A}^{\otimes\bullet};t_\bullet)_{t_\bullet}$ denotes $\bigotimes_{i=1}^{r(\underline{\nu})} \mathrm{HC}_\bullet(\mathcal{A}^{\otimes m_i};t_{m_i})_{t_{m_i}}$ where we use the unique ordering $m_1 \le \cdots \le m_{r(\underline{\nu})}$ of the parts of $\underline{\nu}$ in the non-decreasing order.
- Two-headed arrows denote the quotient maps due to taking the coinvariants by the action of a bigger group.
- The arrows labeled $(\otimes g_{m_i}) \otimes (\otimes g_{n_j})$, EZ, $\xi_\bullet$ and $\xi_\bullet \otimes \xi_\bullet$ mean the images of the corresponding maps after taking the coinvariants as indicated.

These diagrams fit into one big commutative diagram as:

| (5.3) | |
|---|---|
| (5.4) | (5.5) |

In this big diagram the bottom-left half of the perimeter composes to give the LHS of (5.2), and the top-right half composes to give the RHS of (5.2).

The top square in (5.3) commutes tautologically: the right vertical arrow is by definition the unique arrow making this square commute. The bottom square in (5.3) commutes by the associativity of the Eilenberg-Zilber map.

The top square in (5.4) commutes since the maps $\xi_\bullet$ are compatible with the Eilenberg-Zilber maps [GL25, Lemma 5.17]. The bottom square in (5.4) commutes since the symmetrizing maps (4.4) are compatible with the Eilenberg-Zilber maps: the former permute the columns of the Hochschild chains of tensor powers of $\mathcal{A}$, while the latter shuffle the rows while preserving the column positions.

The top square in (5.5) commutes because the induction functor

$$\mathrm{Ind}_{S_m\times S_n}^{S_{m+n}} \colon h\mathcal{P}erf\left(\mathcal{A}^{n+m} \rtimes (S_m \times S_n)\right) \to h\mathcal{P}erf\left(\mathcal{A}^{n+m} \rtimes S_{m+n}\right)$$

restricts on the representables to the fully faithful embedding

$$\mathcal{A}^{n+m} \rtimes (S_m \times S_n) \hookrightarrow \mathcal{A}^{n+m} \rtimes S_{m+n}$$

given by inclusion $S_m \times S_n < S_{m+n}$. The bottom square in (5.5) commutes because for any small DG category $\mathcal{A}$ with strong action of a finite group $G$, the action of

$G$ on $\mathcal{A} \rtimes G$ becomes trivial on the level of Hochschild homology [GL25, Lemma 5.15]. Thus $\mathrm{HH}_\bullet(\mathcal{A} \rtimes G)_G = \mathrm{HH}_\bullet(\mathcal{A} \rtimes G)$ and so (4.4) is the identity map.

We thus conclude that the multiplication $\mu$ is identified by the isomorphisms $\zeta$ with $\mu'$. A similar analysis reduces the statement that the comultiplication $\Delta$ is identified with $\Delta'$ to the commutation of the diagram in [GL25, Lemma 5.18]. □

5.2. **$K$-theoretic parallels and $\lambda$-ring structure.** Let $\mathcal{A}$ be a small DG category. Let $\mathcal{A}$ be equivalent, as an enhanced triangulated category, to the bounded derived category $D(X)$ of coherent sheaves on a smooth projective variety $X$. That is – there is an exact equivalence $D_c(\mathcal{A}) \simeq D(X)$ induced by a Morita equivalence of $\mathcal{A}$ with the standard DG enhancement of $X$ [GKL21, §4.4] [Toë07] [LO10]. By the global version of HKR theorem [Swa96, Cor 2.6], [Kon03, Theorem 4.6.2], [Că05, Theorem 4.9], we have $HH_\bullet(\mathcal{A}) \simeq \bigoplus_{p\geq 0} H^{\bullet-p}(X, \Omega^p)$. When $\Bbbk = \mathbb{C}$, by Poincarè Lemma we further have $HH_\bullet(\mathcal{A}) \simeq H^\bullet(X, \mathbb{C})$ and then by the Chern character isomorphism [AH61, §2.4] we have an isomorphism $HH_\bullet(\mathcal{A}) \simeq K^\bullet(X) \otimes \mathbb{C}$ of $\mathbb{Z}/2$-graded vector spaces, where $K^\bullet(X)$ denotes the topological $K$-theory of $X$.

For any finite group $G$ acting on $X$, in [AS89, Theorem 2] Atiyah and Segal gave a decomposition of the complexified equivariant topological $K$-theory in terms of the fixed point loci of $G$-action:

$$K^\bullet_G(X) \otimes \mathbb{C} \simeq \bigoplus_{[g]\in G} K^\bullet(X_g)_{C(g)} \otimes \mathbb{C}, \tag{5.6}$$

where the sum is taken over the representatives of the conjugacy classes of $G$, $X_g$ denotes the fixed point locus of $g \in G$, and $C(g)$ denotes the centraliser of $g \in G$. The equivariant $K$-theory $K^\bullet_G(X) \otimes \mathbb{C}$ can be viewed as the $K$-theory of the global quotient stack $[X/G]$, and the decomposition (5.6) can be viewed as an earlier $K$-theoretic analogue of the decomposition of the orbifold Hochschild homology (and, more generally, cyclic homology) of $X$ by the second author [Bar03, Theorem 1.1] that this paper generalises to the noncommutative case.

Segal [Seg96, Prop. 1.4] and Wang [Wan00, Theorem 3] used the decomposition (5.6) to identify the total symmetric quotient stack $K$-theory of $X$ with a symmetric algebra:

$$\bigoplus_{n\geq 0} K^\bullet_{S_n}(X^n) \otimes \mathbb{C} \simeq S^* \left( K^\bullet(X) \otimes_{\mathbb{C}} t\mathbb{C}[t] \right). \tag{5.7}$$

This is a $K$-theoretic analogue of our Theorem 4.2 for Hochschild homology. Segal and Wang used it to give $\bigoplus_{n\geq 0} K^\bullet_{S_n}(X^n) \otimes \mathbb{C}$ the structure of the Fock space for the Heisenberg algebra of $K^\bullet(X)$. A Hochschild homology analogue of this Heisenberg algebra action for our noncommutative setting is established in [GL25].

Wang also defined Hopf algebra and $\lambda$-ring structures on $\bigoplus_{n\geq 0} K^\bullet_{S_n}(X^n) \otimes \mathbb{C}$. The Hopf algebra structure [Wan00, §2] is the image under the isomorphism (5.7) of the standard Hopf algebra structure of a symmetric algebra. It is a $K$-theoretic analogue of our Theorem 5.1 for the Hochschild homology. The $\lambda$-ring structure [Wan00, §3] was constructed using isomorphism (5.7) and was shown to be isomorphic to the free $\lambda$-ring generated by $K^\bullet(X) \otimes \mathbb{C}$.

In this section, we investigate whether there is an analogous $\lambda$-ring structure on $\bigoplus_{n\geq 0} \mathrm{HH}(\mathcal{S}^n\mathcal{A})$ in our noncommutative setting.

A $\lambda$-ring is a notion introduced by Grothendieck [Gro71, §I.1-2] axiomatizing the structure defined on the Grothendieck ring $K_0(X)$ of vector bundles on $X$ by

the operation of taking the exterior powers. So a $\lambda$-ring is a unital, commutative ring $R$ equipped with a family of set-theoretic operations $\lambda^i\colon R \to R$ for $i \geq 0$ satisfying certain conditions [Gro71, §I.1-2] [Ber71, Déf. 2.4]. We also refer the reader to [Knu73], [AT69], and more modern [Yau10]. A $\lambda$-ring structure can also be defined in terms of the *Adams operations* $\psi^i\colon R \to R$, and over a field of characteristic 0 these can be any ring homomorphisms satisfying $\psi^1 = \mathrm{Id}$ and $\psi^m \circ \psi^n = \psi^{mn}$, see [Knu73, p. 49] [Wil82, Prop. 1.2].

Given any $E \in \mathbf{Mod}\text{-}\Bbbk$, the symmetric DG algebra $S^*(E \otimes t\Bbbk[t])$ has a natural $\lambda$-ring structure given by

$$\psi^m(vt^n) := vt^{mn} \qquad \forall\, v \in E, n \geq 0. \tag{5.8}$$

With this structure, $S^*(E \otimes t\Bbbk[t])$ is the *free $\lambda$-ring generated by $E$*. It satisfies the universal property that for any other DG $\lambda$-ring $R$ over $\Bbbk$ whose Adams operations are $\mathbf{Mod}\text{-}\Bbbk$-homomorphisms, the DG $\lambda$-ring homomorphisms $S^*(E \otimes t\Bbbk[t]) \to R$ are in bijection with $\mathbf{Mod}\text{-}\Bbbk$-morphisms $E \to R$. The bijection is given by the restriction to $E \otimes t \subset S^*(E \otimes t\Bbbk[t])$.

Thus the isomorphisms $\zeta$ and $\eta$ of Theorem 4.2

$$\bigoplus\nolimits_{n\geq 0} \mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A}) \underset{\eta}{\overset{\zeta}{\underset{\simeq}{\rightleftarrows}}} S^*(\mathrm{HH}_\bullet(\mathcal{A}) \otimes t\Bbbk[t])$$

give the total Hochschild homology $\bigoplus_{n\geq 0} \mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A})$ the structure of a graded $\lambda$-ring with multiplication (5.1) and Adams operations

$$\psi^m\colon\ \mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A}) \to \mathrm{HH}_\bullet(\mathcal{S}^{mn}\mathcal{A}).$$

For $n = 1$ they are given by the composition

$$\mathrm{HC}_\bullet(\mathcal{A}) \xrightarrow{g_m} \mathrm{HC}_\bullet(\mathcal{A}^{\otimes m}; t_m)_{t_m} \xrightarrow{(4.4)} \mathrm{HC}_\bullet(\mathcal{A}^{\otimes m}; t_m) \xrightarrow{\xi_{t_m}} \mathrm{HC}_\bullet(\mathcal{S}^m\mathcal{A}),$$

and analogously for higher $n$.

The $K$-theoretic $\lambda$-ring construction in [Wan00, §3] differs from our as follows. Unlike $HH_\bullet(\mathcal{A})$, the topological $K$-theory $K^\bullet(X) \otimes \mathbb{C}$ has has a natural $\lambda$-ring structure with the multiplication given by the tensor product of vector bundles and $\lambda$-operations – by the exterior powers. The corresponding Adams operations $\psi^m_{K(X)}\colon K^\bullet(X) \to K^\bullet(X)$ are described using the decomposition (5.7) as follows.

Let $V$ be a vector bundle on $X$. Equip $V^{\boxtimes m} := V \boxtimes V \boxtimes \cdots \boxtimes V$ on $X^m$ with the natural action of $S_m$ which acts by permuting the factors. The assignment $V \mapsto V^{\boxtimes m}$ defines a map $\boxtimes m\colon K^\bullet(X) \to K^\bullet_{S_n}(X^n)$. Define

$$\psi^m_{K(X)}\colon K^\bullet(X) \otimes \mathbb{C} \xrightarrow{\boxtimes m} K^\bullet_{S_m}(X^m) \otimes \mathbb{C} \simeq \sum_{\underline{\lambda}\vdash m} \mathrm{Sym}^{\underline{r}(\underline{\lambda})}\, K^\bullet(X) \otimes \mathbb{C}t^{\underline{\lambda}} \twoheadrightarrow K^\bullet(X) \otimes \mathbb{C}t^m, \tag{5.9}$$

where the last map is the projection to the summand given by the long cycle partition $(m)$ of $m$. Each $\psi^m_{K(X)} : K^\bullet(X) \otimes \mathbb{C} \to K^\bullet(X) \otimes \mathbb{C}$ is an automorphism which is identified by the Chern character with the automorphism of $H^*(X, \mathbb{C})$ which acts by the multiplication by $m^k$ on $H^{2k}(X, \mathbb{C})$ and $H^{2k-1}(X, \mathbb{C})$.

The $\lambda$-ring structure constructed in [Wan00, §3] on $\bigoplus_{n\geq 0} K^\bullet_{S_n}(X^n) \otimes \mathbb{C}$ is induced via the isomorphism $\bigoplus_{n\geq 0} K^\bullet_{S_n}(X^n) \otimes \mathbb{C}$ from the $\lambda$-ring structure on the symmetric algebra $S^*\left(K^\bullet(X) \otimes_{\mathbb{C}} t\mathbb{C}[t]\right)$ whose Adams operations are defined by

$$\psi^m(vt^n) := \psi^m_{K(X)}(v)t^{mn} \qquad \forall\, v \in K^\bullet(X), n \geq 0.$$

Comparing this with the canonical Adams operations (5.8) defining the free $\lambda$-ring generated by a graded vector space, we see that the $\lambda$-ring structure constructed on $S^*\left(K^\bullet(X)\otimes_{\mathbb{C}} t\mathbb{C}[t]\right)$ in [Wan00, §3] is not the canonical free one. However, since $\psi^m_{K(X)}$ are automorphisms, it is isomorphic to the canonical one via the $\lambda$-ring automorphism of $S^*\left(K^\bullet(X)\otimes_{\mathbb{C}} t\mathbb{C}[t]\right)$ which sends $K^\bullet(X)t^n$ to $K^\bullet(X)t^n$ via $\psi^n_{K(X)}$.

As $\mathrm{HH}_\bullet(\mathcal{A})$ has no natural $\lambda$-ring structure, we used the canonical Adams operations (5.8) to define $\lambda$-ring structure on $S^*(\mathrm{HH}_\bullet(\mathcal{A})\otimes t\Bbbk[t])$, and hence on $\bigoplus_{n\geq 0}\mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A})$. Thus, in the commutative case, the HKR and Chern character isomorphism $\mathrm{HH}_\bullet(\mathcal{A})\simeq K^\bullet(X)\otimes\mathbb{C}$ doesn't identify our $\lambda$-ring structure on $S^*(\mathrm{HH}_\bullet(\mathcal{A})\otimes t\Bbbk[t])$ with that of [Wan00, §3] on $S^*\left(K^\bullet(X)\otimes_{\mathbb{C}} t\mathbb{C}[t]\right)$. But could it be that the induced $\lambda$-ring structures on $\bigoplus_{n\geq 0}\mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A})$ and $\bigoplus_{n\geq 0}K^\bullet_{S_n}(X^n)\otimes\mathbb{C}$ are identified by the isomorphisms $\mathrm{HH}_\bullet(\mathcal{S}^n\mathcal{A})\simeq K^\bullet_{S_n}(X^n)\otimes\mathbb{C}$ cooked up using the orbifold HKR isomorphism of [ACH19, Theorem 1.15]?

It is not an oversight that our $\lambda$-ring structure is not a noncommutative analogue of the one in [Wan00, §3]. If it were, we would have constructed a noncommutative analogue on $\mathrm{HH}_\bullet(\mathcal{A})$ of the Adams operations $\psi^m_{K(X)}$ on $K^\bullet(X)\otimes\mathbb{C}$. Since the eigenspaces of $\psi^m_{K(X)}$ recover on $K^\bullet(X)\otimes\mathbb{C}$ the cohomological grading of $H^*(X,\mathbb{C})$, we would have constructed a noncommutative Hodge decomposition $H^{i,j}$ on $\mathrm{HH}_\bullet(\mathcal{A})$.

Curiously, though, we now have noncommutative analogues for the classes in $\mathrm{HH}_0(\mathcal{A})$ of the objects of $\mathcal{A}$ of both steps in the definition (5.9). The assignment $V\mapsto V^{\boxtimes m}$ corresponds to sending $a\in\mathcal{A}$ to the $\mathrm{HH}_0$ class of the symmetric idempotent $\sum_{\sigma\in S_m}\sigma$ of the object $a^{\otimes m}$ in $\mathcal{S}^m\mathcal{A}$. This is precisely the Euler class of the $\mathcal{S}^m\mathcal{A}$-module corresponding to the representable $\mathcal{A}^{\otimes m}$-module $a^{\otimes m}$ equipped with the $S_n$-equivariant structure which permutes its factors.

On the other hand, the projection $K^\bullet_{S_m}(X^m)\otimes\mathbb{C}\to K^\bullet(X)\otimes\mathbb{C}$ corresponds in our case to the composition

$$\mathrm{HC}_\bullet(\mathcal{S}^m\mathcal{A})\xrightarrow{\nu}\mathrm{HC}_\bullet(\mathcal{A}^{\otimes m};t_m)_{t_m}\xrightarrow{f_m}\mathrm{HC}_\bullet(\mathcal{A}). \tag{5.10}$$

This sends the symmetric idempotent $\sum_{\sigma\in S_m}\sigma$ of the object $a^{\otimes m}$ first to $\mathrm{Id}_{a^{\otimes m}}$ in $\mathrm{HC}_\bullet(\mathcal{A}^{\otimes m};t_m)_{t_m}$, and then to $\mathrm{Id}_a\in\mathrm{HC}_\bullet(\mathcal{A})$.

Thus replicating in the noncommutative case the steps of the $K$-theoretic Adams operation $\psi_m$ produces a trivial operation on the $\mathrm{HH}_0$ classes of the objects of $\mathcal{A}$. This agrees with our definition of the $\lambda$-structure on $S^*(\mathrm{HH}_\bullet(\mathcal{A})\otimes t\Bbbk[t])$ above. It is interesting whether the noncommutative analogue of the assignment $V\mapsto V^{\boxtimes m}$ can be extended to the whole of $\mathrm{HH}_\bullet(\mathcal{A})$ and whether its composition with the projection (5.10) would be the identity map.

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